INCH-POUND
MIL-PRF-28876F
1 March 2021
SUPERSEDING
MIL-PRF-28876E
4 October 2004

PERFORMANCE SPECIFICATION

CONNECTORS, FIBER OPTIC, CIRCULAR, PLUG AND RECEPTACLE STYLE, MULTIPLE REMOVABLE TERMINI, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers circular, plug and receptacle style, multiple removable termini, fiber optic connectors (see 6.10.3) that are for Department of Defense applications and that are compatible with multiple transmission element cables. Fiber optic connectors specified herein cover a family of general purpose, interconnection hardware providing a variety of compatible optical coupling arrangements. Connector parts specified within this specification include connector shells (see 6.10.11), connector inserts (see 6.10.5), connector insert retention nuts (see 6.10.7), connector backshells (see 6.10.2), and connector dust cover (see 6.10.4).

1.2 Classification. Plug and receptacle styles, as specified in the applicable specification sheet (see 3.1), include straight, wall-mounted (panel-mounted), jamnut-mounted, right angle, and other connector configurations as required for cable system applications.

1.2.1 Termini classification. Termini are classified as described in MIL-PRF-29504.

1.2.2 Connector classification. Connectors fabricated to this specification are classified as follows:

a. Aluminum with conductive plating.

b. Aluminum with non-conductive plating.

1.3 Part or identifying number (PIN).

M	**28876**	**/XX**	**XXXXX**	**X**	**X**
Prefix for Military Specification	Specification Number	Spec Sheet No. (see 3.1)	Four- or five-character alpha-numeric (e.g., shell size designator, insert arrangement, terminus/insert style, key position) (see 3.1)	Backshell designator, when specified (see 3.1)	Class (finish/plating) designator, when specified (see 3.1)

Example: M28876/1A1P1N

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-PRF-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-PRF-29504	-	Termini, Fiber Optic Connector, Removable, General Specification for
MIL-PRF-29504/14	-	Termini, Fiber Optic, Connector, Removable, Environment Resisting, Pin Terminus, Front Release Ceramic Ferrule, (for MIL-C-28876 Connectors)
MIL-PRF-29504/15	-	Termini, Fiber Optic, Connector, Removable, Environment Resisting, Socket Terminus, Front Release, Ceramic Ferrule, (for MIL-C-28876 Connectors)
MIL-PRF-49291/6	-	Fiber, Optical, Type I, Class I, Size IV, Composition A, Wavelength B, Radiation Resistant (Metric)
MIL-PRF-49291/7	-	Fiber, Optical, Type II, Class 5, Size II, Composition A, Wavelength D, Radiation Resistant (Metric)

(See ASSIST database for list of specification sheets).

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1373	-	Screw-Thread, Modified, 60 Degrees Stub, Double
MIL-STD-1678-2	-	Fiber Optic Cabling Systems Requirements and Measurements (Part 2: Optical Measurements) (Part 2 of 6 Parts)
MIL-STD-1678-3	-	Fiber Optic Cabling Systems Requirements and Measurements Physical, Mechanical, Environmental and Material Measurements (Part 3 of 6 Parts)
MIL-STD-1678-4	-	Fiber Optic Cabling Systems Requirements and Measurements (Part 4: Test Sample Configuration and Fabrication Requirements) (Part 4 of 6 Parts)
MIL-STD-2042	-	Fiber Optic Cable Topology Installation Standard Methods for Naval Ships
MIL-STD-2042-6	-	Fiber Optic Cable Topology Installation Standard Methods for Surface Ships and Submarines (Tests) (Part 6 of 7 Parts)
MIL-HDBK-454	-	General Guidelines for Electronic Equipment

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM20995 - Wire, Safety or Lock

(Copies of this document are available online at www.aia-aerospace.org.)

ASTM INTERNATIONAL

ASTM D1149 - Standard Test Methods for Rubber Deterioration-Cracking in an Ozone Controlled Environment

(Copies of this document are available online at www.astm.org.)

ELECTRONIC INDUSTRIES ALLIANCE (EIA)

EIA-359 - Colors for Color Identification and Coding

EIA-364-81 - Combustion Characteristics of Connector Housings, Connector Assemblies and Sockets

(Copies of these documents are available online at www.ihs.org.)

SAE INTERNATIONAL

SAE AMS-C-26074 - Electroless Nickel Coatings

(Copies of this document are available online at www.sae.org.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 62321-7-1 - Determination of Certain Substances in Electrotechnical Products - Part 7-1: Presence of Hexavalent Chromium (Cr(VI)) in Colourless and Coloured Corrosion-Protected Coatings on Metals by the Colorimetric Method

(Copies of this document are available online at www.iec.ch.)

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (ITA)/EIA

TIA-455 - General Requirements for Standard Test Procedure for Fiber Optic Fibers, Cables, Transducers, Sensors, Connecting and Terminating Devices, and Other Fiber Optic Components

TIA-455-2 - Impact Test Measurements for Fiber Optic Devices

TIA-455-6 - FOTP-6 Cable Retention Test Procedure for Fiber Optic Cable Interconnecting Devices

TIA-455-13 - FOTP-13 Visual and Mechanical Inspection of Fiber Optic Components, Devices, and Assemblies

TIA-455-21 - FOTP-21 Mating Durability of Fiber Optic Interconnecting Devices

TIA-455-26 - FOTP-26 Crush Resistance of Fiber Optic Cable Interconnecting Devices

TIA-455-35 - FOTP-35 Fiber Optic Component Dust (Fine Sand) Test

TIA-455-36 - FOTP-36 Twist Test for Fiber Optic Connecting Devices

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TIA/EIA-455-1 - Cable Flexing for Fiber Optic Interconnecting Devices

(Copies of these documents are available online at www.tiaonline.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Fiber optic connectors and accessories furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.5 and 6.3). The provisions of 4.5.4 for retention of qualification are included in this requirement.

3.3 Materials. Materials shall be as specified herein and in the applicable specification sheet. In all cases, materials selected for use shall meet all qualification requirements as specified, and be of a type and quality to assure physical, chemical, and optical compatibility with the requirements of this specification. All materials used shall be nontoxic (see 3.3.5), non-nutrient to fungus (see 3.14.13), and manufactured to good workmanship quality (see 3.10.4). Materials chosen shall be the lightest practicable material suitable for the intended use. Materials shall not interfere with or degrade the terminus (see 6.10.12) cleaning operation and shall be estimated to provide 20-year service.

3.3.1 Connector Parts. Connector shells shall be aluminum. Connector inserts shall be either aluminum or polymer material. Backshells, insert retention nuts, and dust covers shall be either aluminum or a polymer material. Hex nuts provided with a jamnut mounted connector receptacle shall conform to the class designator, except jamnut receptacles with non-conductive plating where the hex nut may be CRES, passivated.

3.3.2 Finish. The resultant finish on all connector parts shall meet the requirements herein and be as follows.

3.3.2.1 Aluminum components. The finish used for external parts shall minimize reflections and shall meet the requirements herein. The finish used for internal parts shall be corrosion resistant.

3.3.2.2 Finish color. Acceptable color on connectors, plug shell, receptacle shell, backshell, and dust cover shall be identified for each finish.

3.3.2.3 Non-conductive finish, aluminum components. The plating of the connectors, plug shell, receptacle shell, backshell, and dust cover shall have a non-conductive coating in accordance with MIL-PRF-8625, type III, class 2, sealed or unsealed, with supplemental impregnated, polytetrafluoroethylene (PTFE) coating. Supplemental PTFE coating shall be the result of impregnating PTFE into the surface asperities of the anodic coating. Buildup per surface for this nonconductive finish shall be between 0.0007 and 0.0012 inch. Color of the non-conductive finish shall be in accordance with 3.3.2.2. A lubricant shall be applied to the connector plug internal coupling nut threads and shall satisfy the lubricant requirements in 3.3.9. A dry film lubricant may be applied to the connector receptacle external mating male threads. For backshells with a base material of die cast aluminum where type III anodize is limited in accordance with MIL-PRF-8625, type II anodize may be used when the procuring activity does not authorize type III. For inserts, a non-conductive coating in accordance with MIL-PRF-8625, type II, type III sealed, or type III unsealed shall be used. Connectors, plug shell, receptacle shell, backshell, and dust cover shall be black as identified in EIA-359. Color shall be minimally reflective.

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3.3.2.4 Conductive plating, aluminum components. When specified (see 3.1), the plating of the connectors, plug shell, receptacle shell, backshell, and dust cover shall be tin-zinc alloy in accordance with SAE AMS2434, type 2, grade A or B with a composition of 70 to 85 percent tin and 15 to 30 percent zinc, 0.0005-inch thick minimum, over electroless nickel in accordance with SAE AMS-C-26074, grade A or B or other similar underplate. Following chromium conversion process, if applicable, and subsequent processing, finish shall be trivalent chromium and shall meet performance requirements. The color shall be gold to bronze, including pale green to olive and brown, or black, and non-reflective (some iridescence permitted but is not preferred). The final finish shall be electrically conductive, shall withstand 500 hours of dynamic salt spray, shall meet the requirements of the hexavalent chromium detection testing, and shall operate in a temperature range of -67 °F to +347 °F (-55 °C to +175 °C). The finish shall be negative for hexavalent chromium (Cr[VI]) when tested in accordance with IEC 62321-7-1, boiling water extraction procedure.

3.3.2.5 CRES components. Unless otherwise specified (see 3.1), all exposed corrosion resistant steel parts shall be passivated and shall be treated to minimize reflections. CRES threads shall be treated in a manner to minimize galling and cold welding with mating parts. CRES parts which may be mated to non-CRES parts shall be treated to minimize scratching or damage to mating components during mating operations.

3.3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3.4 Nonmetallic materials. Nonmetallic materials used in connector parts shall not be degraded by the use of solvents or cleaning agents, nor be degraded at the environmental conditions specified in 3.14.

3.3.5 Toxicity. Materials used in the connectors, backshells, insert retention nuts, and dust covers shall not give off toxic or explosive fumes when exposed to flame. When evaluated in accordance with 4.9, materials used shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.9 and 6.6).

3.3.6 Metals. Metals shall be corrosion resistant or treated to be corrosion resistant. Unless otherwise specified (see 3.1), metals shall be nonmagnetic.

3.3.7 Dissimilar metals. The use of dissimilar metals in intimate contact should be avoided. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided.

3.3.8 Sealing compounds. Sealing compounds shall not flow at the maximum specified storage temperature or exhibit cracking at the minimum specified storage temperature.

3.3.9 Lubricants. Lubricants used in the construction of the connectors shall satisfy the following criteria:

a. Lubricants shall be permanent and shall not require replacement during the lifetime of the connector.

b. Lubricants shall not migrate to the optical interfaces resulting in the degradation of optical performance.

c. Lubricants shall be useful over the environmental conditions specified herein.

d. Lubricants shall not be affected by cleaning solvents.

e. Lubricants shall not corrode or otherwise degrade connector parts.

3.4 Design and construction. Connector parts shall be in accordance with figures A-1 through A-8 and as specified (see 3.1). Connector parts include connector (see 3.5), backshell (see 3.6) and dust cover (see 3.8), and other parts as specified (see 3.1).

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3.4.1 General. Connectors shall be designed to be compatible with optical fibers and cables, as specified (see 3.1).

3.4.2 Seals. Seals shall provide environmental isolation for the optical contact junctions and connector interior parts. Grommets, O-rings, interfacial seals, boots, gaskets, or other sealing devices, as needed by the connector design, shall accomplish their intended purpose and meet all test requirements as specified herein.

3.4.2.1 Optical junction sealing. Optical junctions shall be sealed against moisture and contamination as specified herein.

3.4.2.2 Cable sealing. Connectors shall seal the terminating cables as specified herein.

3.4.3 Interchangeability and interoperability. Connector parts shall be interchangeable and interoperable as specified in 3.4.3.1 and 3.4.3.2.

3.4.3.1 Interchangeability. All connector parts having the same PIN shall be physically and functionally interchangeable without need for modification of such items or of the mating equipment.

3.4.3.2 Interoperability. All connectors of the same PIN shall be interoperable. Upon qualification of the first manufacturer, all subsequent manufacturers shall provide proof of interoperability with each qualified manufacturer as specified in 4.8. The connectors shall meet the requirements of 3.11.1, 3.12.3, and 3.12.4 for each specified interoperability condition (see 4.8).

3.5 Connectors parts.

3.5.1 Shells. The connector shells shall house the connector insert, and the insert retention nut or backshell shall retain the connector insert. The plug to receptacle connections shall be environmentally and water sealed.

3.5.1.1 Plugs. Plugs shall be of the inline (straight), 45-degree, and 90-degree type as specified (see 3.1).

3.5.1.2 Receptacles. Receptacles shall be of the wall (panel) mount, jamnut, or inline types as specified (see 3.1).

3.5.1.3 Engagement of connectors. Counterpart connectors of any arrangement and accessories shall be capable of being fully engaged and disengaged without the use of tools, except shell size 23.

3.5.1.4 Coupling mechanism. Coupling rings of the connectors shall be knurled and designed so that plug and receptacle optical termini shall approach or recede from each other as the coupling mechanism is respectively tightened by clockwise rotation or loosened in the counterclockwise direction as viewed from the rear of the plug connector. The coupling mechanism shall be captive on the plug to mate with the receptacle shell. Coupling ring and coupling screw threads shall be in accordance with MIL-STD-1373, as shown on figures A-4 and A-5, and as specified (see 3.1). Lubricated coupling threads shall meet all of the requirements specified herein (see 3.3.9).

3.5.1.5 Plug and receptacle shell polarization (keying). Polarization keying shall be incorporated in the shells of plugs and receptacles to assure correct alignment of the inserts before mating is permitted. The polarization shall be accomplished by integral keys in the shells (see figure A-3). The keying shall be designed to prevent physical contact of the mating optical termini, or of the termini with the insert surface of the counterpart connector until the keyways are properly aligned for engagement and coupling mechanisms are engaged.

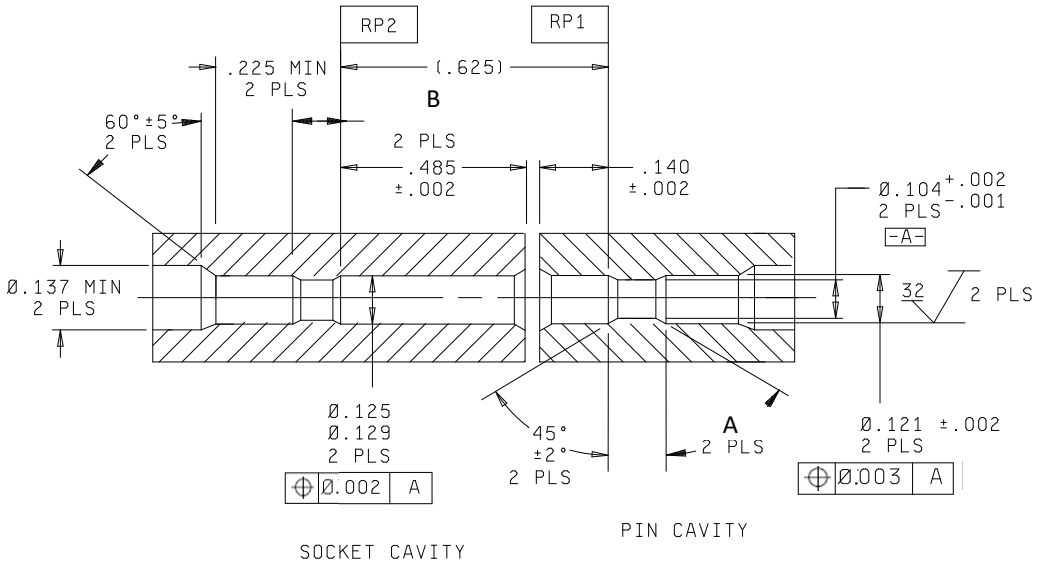
3.5.1.6 Safety wiring. Threaded plug coupling rings and the hex nuts on jam-nut receptacles shall be designed for safety wiring. For plug coupling rings, at least two equally spaced holes shall be provided for shell size 15 and smaller, and at least three equally spaced holes shall be provided for connector shell sizes 17 and larger. For hex nuts, at least three holes at alternating hex points shall be provided. The exact orientation and location of holes is not specified. The typical location shall be as specified (see 3.1). These holes shall be sufficient to accommodate 0.032-inch diameter wire in accordance with NASM20995.

3.5.2 Termini. Removable, environment resisting fiber optic termini shall be used with these connectors for interconnecting fiber optic cables. The termini are not supplied with connectors acquired to this specification. Termini, for use with the connectors specified herein, shall be in accordance with MIL-PRF-29504 (see 3.1).

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3.5.3 Inserts. Inserts shall be keyed and secured to prevent rotation within the connector shell. Inserts shall be of a one-piece or two-piece construction and shall be such that they will not crack, chip, or break in normal service or assembly. Inserts glued or bonded together shall not be used. The insert dimensions shall be as specified (see 3.1). The insert hole configuration shall conform to the dimensions shown on figure 1.

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	Dimension A	Dimension B
Default dimension	60±5º	0.097±0.004
Alternate dimension	70±1.5º	0.099±0.022

NOTES:

1. All dimensions are in inches unless otherwise noted.
2. Mating cavities to be aligned Ø.010 Ⓢ and insert faces abutted when connectors are mated.
3. The following dimensions should be used as a reference:
   M29504/14 ceramic tip pin termini protrude 0.378 from R.P.1
   M29504/15 ceramic tip socket termini with alignment sleeve protrude 0.423 from R.P.2

Inch	Millimeters	Inch	Millimeters
0.001	0.03	0.137	3.48
0.002	0.05	0.140	3.56
0.003	0.08	0.225	5.72
0.004	0.10	0.342	8.69
0.010	0.25	0.378	9.60
0.097	2.46	0.423	10.74
0.104	2.64	0.428	10.87
0.121	3.07	0.485	12.32
0.125	3.18	0.625	15.88
0.129	3.28	--	--

FIGURE 1. Termini cavity dimensions.

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3.5.3.1 Number of termini, arrangement, and spacing. The insert pattern (i.e., the number of termini) and their arrangement and spacing shall be as specified in Appendix B. Every terminus position shall accept either optical or dummy termini.

3.5.3.2 Terminus insertion and removal methods. Optical terminus insertion shall be accomplished by inserting the terminus, using a terminus insertion tool, into the rear of the connector insert. A means for locking the terminus in place shall be provided. Optical terminus removal shall be accomplished by inserting the terminus removal tool into the front of the connector and by forcing the terminus out the rear of the connector. The individual termini shall be positively retained in the connector when installed with the terminus insertion tool and shall be capable of being removed without terminus or insert damage when using the terminus removal tool. Requirements for these tools shall be as specified (see 3.1).

3.5.4 Assembly methods. Connector assembly shall be accomplished in accordance with MIL-STD-2042.

3.6 Backshells and insert retention nuts.

3.6.1 Backshells. Backshells shall be as specified (see 3.1). The backshells shall be provided with cable strain relief as specified (see 3.1). The backshells shall be free of any sharp edges or other configurations that could cause damage to optical fibers extending through them.

3.6.2 Insert retention nuts. Insert retention nuts shall be as specified (see 3.1). The insert retention nuts shall be provided without cable strain relief as specified (see 3.1). The insert retention nuts shall be free of any sharp edges or other configurations that could cause damage to optical fibers extending through them.

3.7 Protective caps or covers. All optical connectors (plugs and receptacles) shall be provided with throwaway caps or covers. The cap or cover shall be free of mold release, lubricants, or any other contaminants.

3.8 Dust covers. Dust covers shall be as specified (see 3.1). The dust covers shall be free of any sharp edges or any other configurations that could cause damage to the optical termini.

3.9 Tools. Tools used to terminate connectors onto cables shall be as specified (see 3.1).

3.10 Visual and mechanical.

3.10.1 Size. When examined in accordance with 4.6.2.1, the dimensions and dimensional tolerances for the connector parts (see 3.1 and 3.5 through 3.8) shall be as specified (see 3.1).

3.10.2 Weight. When tested in accordance with 4.7.2.2, the weight of the connector parts shall be as specified (see 3.1).

3.10.3 Identification marking (see 4.7.2.3). Identification marking shall be as specified in 3.10.3.1 and 3.10.3.2. Markings shall be legible and permanent. Markings shall be legible to the extent that none are missing (in whole or in part), faded, blurred, smeared, or shifted (dislodged), and shall be readily readable. Contrast between characters and surface shall be good. Markings shall be permanent to the extent of withstanding cleaning procedures and of withstanding environmental and mechanical performance tests conducted.

3.10.3.1 Connectors, backshells, and dust covers. Connectors, backshells, and dust covers shall be identified with markings that are permanent, clearly visible, and legible. Identification marking shall include the PIN and either the manufacturer's CAGE code, name, or logo. Connectors shall also be marked with a yellow band in accordance with EIA-359 or the phrase "FIBER OPTICS", as specified (see 3.1). Connectors, backshells, and dust covers shall be marked with a date code "yy/ww" or "YYWW" along with the phrase "HA" (to designate Hard Anodize PTFE finish) for parts with Hard Anodize PTFE finish (see 3.3.2.3).

3.10.3.2 Inserts. Marking shall correspond between mating inserts and shall be as specified in Appendix B. Raised or depressed characters shall not be used on mating faces. Terminus locations shall be designated by identifiable characters on the front and rear faces of the insert or the insert assembly. Character position and arrangement shall assure appropriate terminus cavity identification.

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3.10.3.3 JAN and J marking. The United States Government has adopted and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to and meet all the requirements of specifications. Accordingly, items acquired to and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets, the manufacturer shall remove completely the military part number and the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 6,069,576 for the certification mark "JAN" and Registration Number 2,577,735 for the certification mark "J".

3.10.4 Workmanship. All details of workmanship shall be as specified herein when examined in accordance with 4.7.4. Connector parts, including backshells and dust covers, shall be dimensionally uniform and free of manufacturing flaws that would degrade performance, inhibit proper connection to interfacing elements, and otherwise yield an inferior product. When examined in accordance with 4.7.4, the following shall not be permitted:

a. Loose termini, inserts, or other connector parts which adversely affect the environmental sealing, or degrade optical termini alignment.

b. Peeling or chipping of plating or finish, galling of mating parts indicating excessive wear, nicks, burrs, or other substandard connector surface blemishes.

3.10.5 Screw threads. When tested in accordance with 4.7.2.5, slight out-of-roundness beyond the tolerances specified shall be considered acceptable if the threads can be checked without forcing the thread gauges. Screw threads may be relieved, provided the relief does not interfere with proper performance of the screw threads.

3.10.6 Maintainability. The connectors shall require no preventive maintenance.

3.11 Optical performance. The optical performance requirements of 3.11.1 through 3.11.5 shall be used to monitor effects of the inspection requirements specified in 3.12 and 3.13 as specified in 4.5.

3.11.1 Insertion loss. The initial insertion loss (see 6.10.8) and the insertion loss verified at any time during testing for a mated pair of connectors shall be not greater than the values specified in table I for each of the mated measurements. The insertion loss shall be measured for 10 mates and demates, except for shell size 23 which shall be measured for 5 mates and demates (see 4.7.3.1).

TABLE I. Insertion loss.

Fiber type (MIL-PRF-49291)	Terminus (MIL-PRF-29504)	Fiber style	Initial insertion loss (dB)	Insertion loss verification (dB)
/6	/14	62.5/125	0.75	1.25
/6	/15	62.5/125	0.75	1.25
/7	/14	SM	0.75	1.25
/7	/15	SM	0.75	1.25

3.11.2 Discontinuities. When measured in accordance with 4.7.3.2, no discontinuity shall occur. For multimode termini, a discontinuity is considered to be a reduction of optical transmittance of 0.5 decibels (dB) or more for a duration of 50 microseconds or more. For single mode termini, a discontinuity is considered to be a reduction of optical transmittance of 0.5 dB or more for a duration of 50 microseconds or more (during vibration) or 100 milliseconds or more (during shock).

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3.11.3 Crosstalk. When connectors with three or more channels are tested in accordance with 4.7.3.3, the signal power levels, or sum of levels of the passive channel or channels, shall be below the signal level of the active channel by at least 60 dB.

3.11.4 Change in optical transmittance. When tested in accordance with 4.7.3.4, the change in optical transmittance shall be less than 0.5 dB.

3.11.5 Return loss. When measured in accordance with 4.7.3.5, the return loss of a standard polish multimode or single mode optical terminus shall be not less than 30 dB. The return loss of an enhanced polish single mode optical terminus shall be not less than 40 dB.

3.12 Functional requirements.

3.12.1 Insert retention radial strength. When tested in accordance with 4.7.4.1, connector inserts shall withstand the clockwise and counterclockwise radial torque specified in table II for a minimum period of one minute. No rotational displacement shall be observed between the inserts and their shell body during or after the test exposure.

TABLE II. Insert retention radial strength.

Connector shell size	Minimum radial torque inch-pounds (N m)
11	15 (1.7)
13	20 (2.3)
15	25 (2.8)
23	30 (3.4)

3.12.2 Insert retention axial strength. When tested in accordance with 4.7.4.2, connector inserts shall withstand an applied minimum pressure of 100 pounds per square inch (0.69 Megapascal [MPa]) in both the forward direction and the backward direction for a minimum period of 1 minute without cracking, breaking, or being dislocated from their normal positions in the connector shell. No axial displacement detrimental to performance shall be observed between the inserts and their shell body during or after the test exposure.

3.12.3 Terminus insertion and removal forces. Connectors shall be tested in accordance with 4.7.4.3. The terminus insertion force and the force required to remove unlocked termini shall not exceed 22 pounds (98 Newtons).

3.12.4 Terminus retention force. When tested in accordance with 4.7.4.4 and subjected to axial loads of 22 pounds (98 Newtons) termini shall be retained in their inserts and meet the requirements of 3.12.4.1 and 3.12.4.2, as applicable.

3.12.4.1 Socket termini. Socket termini shall be retained in their inserts and axial displacements of the termini shall not exceed 0.015 inch (0.38 millimeter) from the pre-load position while under the maximum specified axial load.

3.12.4.2 Pin termini. Pin termini (being spring loaded) may deflect more than 0.015 inch (0.38 millimeter) while under axial load, but must return to within 0.015 inch (0.38 millimeter) axial displacement from the pre-load position after the axial load is removed.

3.12.5 Maintenance aging. Connectors with removable termini shall be tested in accordance with 4.7.4.5. Connectors shall show no visible evidence of wear or deformation which may degrade their ability to perform as specified. The terminus insertion and removal forces requirement of 22 pounds (98 Newtons) (see 3.12.3) shall be met.

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3.12.6 Connector coupling engagement and disengagement torque. When tested in accordance with 4.7.4.6, the maximum coupling ring and coupling screw engagement and disengagement torques shall be as specified in table III.

TABLE III. Connector coupling engagement and disengagement torques.

Connector shell size	Connector coupling engagement and disengagement torque, inch-pounds (N m)
11	15 (1.7)
13	15 (1.7)
15	25 (2.8)
23	50 (5.7)

3.12.7 Backshell and insert retention nut attachment. When tested in accordance with 4.7.4.7, the minimum backshell or insert retention nut disengagement torque shall be as specified in table III. No evidence of excessive thread binding, seal pinching, or any contamination buildup shall be observed. This requirement is not applicable for connectors with integral backshells.

3.13 Mechanical requirements.

3.13.1 Cable pull out force (connectors with backshells only). When tested in accordance with 4.7.5.1, the minimum cable-to-backshell pullout strength shall be 162 pounds (720.3 Newtons). There shall be no evidence of cable jacket damage, cable strain relief failure, cable to backshell seal damage, distortion or bending of metallic connector parts, or cable disengagement from the cable strain relief. The connector shall meet the requirements of 3.11.4 during and after the test.

3.13.2 External bending moment (connectors with backshells only). When tested in accordance with 4.7.5.2, connectors and backshells shall exhibit no visible evidence of damage that may degrade their ability to perform as specified (see 3.1).

3.13.3 Cable seal flexing. When tested in accordance with 4.7.5.3, connector strain relief mechanisms shall prevent loss of environmental sealing or other damage which may impair the connector operation.

3.13.4 Twist. When tested in accordance with 4.7.5.4, connector seals shall not be rendered inoperable nor shall any other connector damage occur. The requirements of 3.11.4 shall be met during and after the test.

3.13.5 Mating durability. When tested in accordance with 4.7.5.5, mating connectors shall show no evidence of mechanical defects detrimental to the operation of the connector. The requirements of 3.11.4 shall be met during and after the test.

3.13.6 Impact. When tested in accordance with 4.7.5.6, connectors shall not be visibly damaged or otherwise rendered unfit for operational use. The requirements of 3.11.4 shall be met after the test.

3.13.7 Crush. When tested in accordance with 4.7.5.7, connectors shall show no evidence of inability to mate or unmate, broken parts, loss of optical continuity, or damage to shells, backshells, or dust covers. The requirements of 3.11.4 shall be met during and after the test.

3.14 Environmental requirements.

3.14.1 Temperature ranges. The connectors shall meet all requirements specified (see 3.1), during the specified operating environments and after the specified storage environments. The operating temperature range and storage temperature range shall be as shown in table IV, as specified (see 3.1).

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TABLE IV. Temperature ranges.

Operating extremes, °F (°C)	Storage extremes, °F (°C)	Non-operating extremes, °F (°C)
-18.4 to +149 (-28 to +65)	-40 to +158 (-40 to +70)	-40 to +158 (-40 to +70)

3.14.2 Thermal shock. When tested in accordance with 4.7.6.1, connectors shall show no evidence of mechanical damage, loosening of component parts, separation of bonded surfaces, or other damage detrimental to the operation of the connector. The connector shall meet the requirements of 3.11.4 after the test.

3.14.3 Temperature/humidity cycling. When tested in accordance with 4.7.6.2, connector parts shall not swell or otherwise degrade such that connector performance is impaired. The requirements of 3.11.4 shall be met during and after the test.

3.14.4 Temperature cycling. When tested in accordance with 4.7.6.3, a post-test visual examination of the test connectors shall reveal no evidence of connector part dimensional change, no leakage of waterproofing compounds or other apparent loss of sealing capability, no surface or identification marking impairment, no coupling-thread binding or other evidence of mating or unmating incapability, and no other damage detrimental to the operation of the connector. The requirements of 3.11.4 shall be met during and after the test.

3.14.5 Life aging. When tested in accordance with 4.7.6.4, connectors subjected to these specified accelerated aging exposures shall not exhibit visual evidence of dimensional change, opening of seals, cracking or crazing of components or finishes, identification marking impairment, fusion or seizure of mating parts, leakage of waterproofing compounds, or other effects detrimental to the operation of the connector. The connector shall meet the requirements of 3.11.4 after the test. Connectors with dielectric inserts shall meet the insert retention radial strength (see 3.12.1) and insert retention axial strength (see 3.12.2) requirements after aging.

3.14.6 Freezing water. When tested in accordance with 4.7.6.5, connectors shall not be physically damaged. The connectors shall meet the optical requirements specified in 3.11.4 during and after the test.

3.14.7 Sand and dust. When tested in accordance with 4.7.6.6, the connectors shall show no evidence of physical damage which will adversely affect the operation of the connector. The change in optical transmittance requirements of 3.11.4 shall be met during and after the test, and coupling torques requirements of 3.12.6 shall be met after the test.

3.14.8 Terminus cleaning. After cleaning the terminus in accordance with 4.7.6.7, termini removed from the connector shall meet the marking requirements of 3.10.3.2 (see 4.7.6.7) and termini within the connector shall meet the change in optical transmittance requirements of 3.11.4 after the test.

3.14.9 Electromagnetic effects. When tested in accordance with 4.7.6.8, the propagated radio frequency (RF) attenuation of the connector shall be not less than 58 dB for frequencies not greater than 10,000 MHz.

3.14.10 Fluid immersion. When tested in accordance with 4.7.6.9, visual examination of the test connector shall reveal no swelling or softening of material, no loss of sealing capability or identification marking, and no discoloration or other effects detrimental to the intended use of these connectors.

3.14.11 Salt spray (corrosion). When tested in accordance with 4.7.6.10, no visible evidence of salt penetration into the connector sealed area shall be observed. No corrosive effects shall be seen on the external connector parts that would be detrimental to the operation of the connector. For conductive plating/finish class only, the after conditioning requirements of 3.11.18 shall be met after salt spray exposure.

3.14.12 Flammability. When tested in accordance with 4.7.6.11, the mated cable-connector assembly shall meet the optical requirements of 3.11.4. The unmated connector assembly shall not exceed a combined flame and afterglow extinguishing time of 30 seconds after removal of the applied flame. There shall be no dripping that will cause the flammable material to ignite and there shall be no violent burning or explosive type fire.

3.14.13 Fungus resistance. When tested in accordance with 4.7.6.12, polymeric connector materials shall show sparse or very restricted microbial growth and reproduction with minor or inhibited substrate utilization. There shall be little or no chemical, physical, or structural change detectable.

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3.14.14 Ozone exposure. When tested in accordance with 4.7.6.13, seals shall show no evidence of excessive swelling or embrittlement which may degrade environmental isolation.

3.14.15 Vibration. When tested in accordance with 4.7.6.14, the requirements of 3.11.2 shall be met during the test and the requirements of 3.11.4 shall be met after the test. Connectors shall not exhibit relative motion between parts or other damage which can produce physical distortion or wear that may result in fatigue of the mechanical parts or failure of connector operation. During the swept sine test, connectors shall not exhibit visual evidence of loosening of parts for frequencies less than 75 Hz.

3.14.16 Shock. When tested in accordance with 4.7.6.15, connectors shall not be damaged and there shall be no loosening of parts. The requirements of 3.11.2 shall be met during the test and the requirements of 3.11.4 shall be met after the test.

3.14.17 Water pressure. When tested in accordance with 4.7.6.16, visual inspection of the test connector shall reveal no penetration of water into the sealed region of the mated connector. The requirements of 3.11.4 shall be met after the test.

3.14.18 Receptacle (conductive)-to-bulkhead conductivity. When tested in accordance with 4.7.6.17, the maximum measured initial voltage potential drop between the conductive receptacle and bulkhead shall be 2.5 millivolts. The maximum value shall not exceed 5.0 millivolts after conditioning (see 3.14.11).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.5).

b. Conformance inspection (see 4.6).

4.2 Verification program. Requirements for the verification program shall be as identified in the qualification instructions (see 6.3.3).

4.2.1 Assembly plants. Assembly plants must be listed on, or approved for listing on, the applicable qualified products list. The assembly plant shall use only piece parts supplied by the qualified connector manufacturer. No testing other than visual examination is required of certified piece parts obtained from the qualified connector manufacturer, except when there is cause for rejection. All assemblies produced at the assembly plant shall be subjected to examination of the product to ensure that the assembly process conforms with that established at the qualified manufacturing plant. Conformance requirements shall be the same as required for the qualified connector manufacturer.

4.3 Test equipment and inspection facilities. Requirements for test equipment and inspection facilities shall be as identified in the qualification instructions (see 6.3.4).

4.4 Inspection conditions. Unless otherwise specified (see 3.1), all inspections shall be performed in accordance with the test conditions specified in TIA-455 or as specified herein.

4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production. This inspection shall consist of performing the inspections and optical tests specified in table V, in the sequence shown therein, on the qualification test samples specified in 4.5.1.

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TABLE V. Qualification inspection. [1/]

Inspection	Part applicability			Requirement	Verification
	Connector	Backshell	Dust covers		
Group I (six mated pairs)					
Interoperability [2/]	X	X	--	3.4.3.2	4.8
Visual and mechanical			--	3.10	4.7.2
Size	X	X	X	3.10.1	4.6.2.1
Weight	X	X	X	3.10.2	4.7.2.2
Identification marking	X	X	X	3.10.3	4.7.2.3
Workmanship	X	X	X	3.10.4	4.7.2.4
Screw threads	X	--	X	3.10.5	4.7.2.5
Functional [3/]	--	--	--	3.12	4.7.4
Insert retention radial strength	X	--	--	3.12.1	4.7.4.1
Insert retention axial strength	X	X	--	3.12.2	4.7.4.2
Terminus retention force	X	--	--	3.12.4	4.7.4.4
Maintenance aging	X	--	--	3.12.5	4.7.4.5
Connector coupling engagement and disengagement torque	X	--	X	3.12.6	4.7.4.6
Backshell and insert retention nut attachment	X	X	--	3.12.7	4.7.4.7
Optical	--	--	--	3.11	4.7.3
Insertion loss (initial)	X	X	--	3.11.1	4.7.3.1
Return loss	X	X	--	3.11.5	4.7.3.5
Crosstalk	X	--	--	3.11.3	4.7.3.3
Group II (two mated pairs)					
Cable pull out force	X	X	--	3.13.1	4.7.5.1
External bending moment	X	X	--	3.13.2	4.7.5.2
Cable seal flexing	X	X	--	3.13.3	4.7.5.3
Twist	X	X	--	3.13.4	4.7.5.4
Mating durability	X	--	--	3.13.5	4.7.5.5
Return loss	X	--	--	3.11.5	4.7.3.5
Impact	X	X	X	3.13.6	4.7.5.6
Crush	X	X	--	3.13.7	4.7.5.7
Insertion loss verification	X	--	--	3.11.1	4.7.3.1
Vibration	X	X	--	3.14.15	4.7.6.14
Shock	X	X	--	3.14.16	4.7.6.15
Water pressure	X	X	--	3.14.17	4.7.6.16

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TABLE V. Qualification inspection – Continued. [1/]

Inspection	Part applicability			Requirement	Verification
	Connector	Backshell	Dust covers		
Group III (two mated pairs)					
Thermal shock	X	X	--	3.14.2	4.7.6.1
Temperature/humidity cycling	X	X	--	3.14.3	4.7.6.2
Temperature cycling	X	X	--	3.14.4	4.7.6.3
Life aging	X	X	--	3.14.5	4.7.6.4
Insert retention radial strength [4/]	X	--	--	3.12.1	4.7.4.1
Insert retention axial strength [4/]	X	--	--	3.12.2	4.7.4.2
Freezing water	X	X	--	3.14.6	4.7.6.2
Insertion loss verification	X	X	--	3.11.1	4.7.3.1
Sand and dust	X	X	X	3.14.7	4.7.6.6
Connector coupling engagement and disengagement torque	X	--	X	3.12.6	4.7.4.6
Terminus cleaning	X	--	--	3.14.8	4.7.6.7
Identification marking	X	X	X	3.10.3	4.7.2.3
Group IV (two mated pairs and parts)					
Receptacle (conductive)-to-bulkhead conductivity (class T only)	X	--	--	3.14.18	4.7.6.17
Electromagnetic effects (two mated pairs) [4/]	X	X	X	3.14.9	4.7.6.8
Fluid immersion (assemblies and parts [5/], see 4.7.6.9)	X	X	--	3.14.10	4.7.6.9
Salt spray (two mated pairs)	X	X	X	3.14.11	4.7.6.10
Flammability (one mated pair)	X	X	--	3.14.12	4.7.6.11
Fungus resistance (parts [5/])	X	X	X	3.14.13	4.7.6.12
Ozone exposure (parts [5/])	X	X	X	3.14.14	4.7.6.13

NOTES:

[1/] Group I mated pairs shall be used for groups II, III, and IV.

[2/] Separate samples shall be used for interoperability testing. Connector interoperability samples shall be fully populated with termini on single fiber cable and configured without backshells. Backshell interoperability samples shall be constructed with terminated cables that fully populate the connector of the shell size under test.

[3/] Six additional samples may be prepared for functional tests only. Termini used for functional test may be epoxy filled and polished termini, fiber not required.

[4/] Shall be performed only for connector samples utilizing non-metallic inserts.

[5/] Polymeric parts from one mated pair.

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4.5.1 Test samples. Fiber optic connector parts complying with the specified requirements (see 3.1) shall be selected for qualification testing from units produced on typical manufacturing lines. The manufacturer shall provide a counterpart connector for each connector subjected to qualifying tests requiring mating assemblies. The counterpart connectors provided for this purpose shall be new, previously qualified connectors or new connectors submitted for qualification testing. For those tests specifying the use of mated connectors, optical and mechanical test assessments shall be made using the assigned counterpart connector for those test measurements as required.

4.5.1.1 Sample size. Six mated pairs and one wall-mounted receptacle (without backshell) shall be submitted for qualification testing. Six additional samples may be prepared for functional tests only. These connectors shall consist of MIL-PRF-28876/2 wall-mounted (panel-mounted) receptacles with straight backshells and MIL-PRF-28876/7 plugs with straight backshells or MIL-PRF-28876/1 wall-mounted (panel-mounted) receptacles with MIL-PRF-28876/27 backshells and MIL-PRF-28876/6 plugs with MIL-PRF-28876/27 backshells as specified (see 3.1).

4.5.1.2 Sample preparation. Unless otherwise specified (see 3.1 and table V), connectors shall be fully assembled into cable-connector assemblies using the types of cable specified in the applicable connector specification sheet (see 3.1), and the termination procedure shall be in accordance with MIL-STD-1678-4, Requirement 4102. Unless otherwise specified (see 3.1 and table V), connector termini shall be optically finished with termini properly seated within their inserts. For mated connectors, full sealing capability shall be provided as specified. Connectors shall be provided with backshell, and attached to an appropriate length of the specified cable type.

4.5.1.2.1 Termini. The minimum number of active (optically functional) termini required for testing for each shell size shall be as follows:

a. A minimum of two termini shall be active in connectors of shell size 11.

b. A minimum of four termini shall be active in connectors of shell sizes 13 and 15.

c. A minimum of eight termini shall be active in connectors of shell size 23.

d. For shell sizes 15 and 23, the active termini shall be placed at different locations in each of the six samples. Non-active termini shall be, at minimum, M29504/14 or M29504/15 termini with cured epoxy.

4.5.2 Inspection routine. Connector parts shall be tested in accordance with the sequence of table V. Manufacturers desiring qualification for only backshells or dust covers shall, at a minimum, test connectors as specified in 4.5.1 in those tests indicated in table V. Group II, III, and IV testing may be conducted simultaneously.

4.5.2.1 Qualification of modified designs. Manufacturer(s) shall notify the qualifying activity of any design changes. Qualification inspection of items which contain design changes from previously qualified items may be limited to a subset of the qualification inspections identified in 4.5.2. Qualification inspection reductions will be determined by the qualifying activity based on the extensiveness of the design changes and the anticipated effects of those changes on the item performance.

4.5.3 Failures. One or more failures shall be cause for refusal to grant qualification approval.

4.5.4 Retention of qualification. To retain qualification, the manufacturer shall verify in coordination with the qualifying activity the capability of manufacturing products which meet the performance requirements of this specification. Refer to the qualifying activity for the guidelines necessary to retain qualification with respect to this particular specification. The manufacturer shall immediately notify the qualifying activity at any time that the inspection data indicates failure of the qualified product to meet the performance requirements of this specification.

4.5.4.1 Retention of qualification for group A inspections. At intervals of not more than 12 months after initial qualification and every 12-month period thereafter, group A inspections for retention of qualification shall be performed on interconnection boxes and shall include the examinations and tests of table VI. Sampling for group A inspection shall be as specified in 4.6.1.2.1. Manufacturers shall certify that there have been no material changes since initial qualification

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4.5.4.2 Retention of qualification for group B inspections. At intervals of not more than 24 months after initial qualification, and every 24-month period thereafter, group B inspections for retention of qualification shall be performed and shall include the examinations and tests of table VI. Sampling for group B inspection shall be as specified in 4.6.2.1. Manufacturers shall certify that there have been no material changes since initial qualification.

4.5.4.3 Retention of qualification for group C inspections. At intervals of not more than 60 months after initial qualification and every 60-month period thereafter, Group C inspections for retention of qualification shall be performed and shall include the examinations and tests of table VI. Periodic inspection shall consist of Group C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.6.3.5), delivery of products which have passed Group A and Group B inspections shall not be delayed pending the results of periodic inspection.

TABLE VI. Conformance inspections (groups A, B, and C).

Inspection	Component applicability groups A, B, and C			Number of components/ number of connector mated pair 1/
	Connector	Backshell	Dust cover	
Component only				
Group I				
Visual and mechanical				
Size	A	A	A	2/, 3/
Weight	A	A	A	2/, 3/
Identification marking	A	A	A	3/
Workmanship	A	A	A	3/
Screw threads	A	A	A	2/, 3/
Functional				
Backshell and insert retention nut attachment	A	A	--	3/
Insert retention radial strength	B	--	--	Four sample units
Insert retention axial strength	B	--	--	Four sample units
Terminus insertion and removal forces	B	--	--	Four sample units
Terminus retention force	B	--	--	Four sample units
Connection coupling engagement and disengagement torque	B	B	B	Four sample units 4/, 5/
Group II				
Cable pullout force	--	B	--	One test sample - # 1 7/
Water pressure	--	B	--	One test sample - # 1 7/
Group IV				
Receptacle (conductive)-to-bulkhead conductivity (conductive finish/plating only)	C	--	--	One sample unit 10/
Salt spray	C	C	C	One sample unit 10/

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TABLE VI. Conformance inspections (Groups A, B and C) – Continued.

Inspection	Component applicability groups A, B, and C			Number of components/ number of connector mated pair [1/]
	Connector	Backshell	Dust cover	
Component only				
Connector mated pair terminated with single fiber cables and no backshell				
Group I				
Optical				
Insertion loss (initial)	B	--	--	Two- connector mated pair - # 1, 2 [6/]
Return Loss (SM)	B	--	--	Two- connector mated pair - # 1, 2 [6/]
Group III				
Temperature cycling	B	--	--	Two- connector mated pair - # 1, 2 [6/]
Connector mated pair terminated with multiple fiber cable and backshell				
Group I				
Optical				
Insertion loss	C	C	--	Three connector mated pair - # 1, 2, 3 [8/]
Return loss (SM)	C	C	--	Three connector mated pair - # 1, 2, 3 [8/]
Group II				
Mating durability	C	--	--	Three connector mated pair - # 1, 2 [8/]
Return loss (SM)	C	--	--	Two connector mated pair - # 1, 2 [8/]
Shock	C	C	--	Two connector mated pair - # 1, 2 [8/]
Group III				
Temperature humidity cycling	C	C	--	One connector mated pair - # 3 [8/]
Life aging	C	--	--	One connector mated pair - # 3 [8/, 9/]
Insert retention radial strength	C	--	--	One connector mated pair - # 3 [8/, 9/]
Insert retention axial strength	C	--	--	One connector mated pair - # 3 [8/, 9/]

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TABLE VI. Conformance inspections (Groups A, B and C) – Continued.

NOTES:

1/ Selection is based on each component specification sheet being offered with a few exceptions. For connector specification sheets, this may encompass multiple specification sheets if the same connector (either connector plug or connector receptacle) is used for the connector (stand-alone) and for integral connectors with backshells. For backshell specification sheets, this may encompass three specification sheets if the same backshell design and parts are used for straight, 45-degree and 90-degree backshells.

2/ In-process inspections or controls may be used to verify conformance to these requirements.

3/ For sample size, see group A conformance requirements.

4/ Shall be fully populated with terminated M29504/14 and M29504/15 termini inserted into the insert.

5/ Group B backshell sample units for group I: two straight backshells, one 45-degree backshell, one 90-degree backshell.

6/ Group B connector mated pair configuration for groups I and III. Two-connector mated pair consisting of a connector plug mated to a connector receptacle fully populated with termini on single fiber cable.

7/ Group B backshell test sample configuration for group II. One test sample shall consist of a connector plug dust cover, connector plug, straight backshell and a minimum 1 meter of multiple fiber cable existing the backshell. The multiple-fiber cable end inside the backshell shall be strain relieved with backshell seals in place. The multiple-fiber cable end outside the backshell shall be adequately sealed. For purposes of securing to a fixture for cable pullout, a connector receptacle may be used in lieu of the connector plug dust cover. No optical measurements are required for these backshell group B tests.

8/ Group C connector mated pair configuration:

Connector mated pair # 1: connector plug with 45-degree backshell and connector receptacle with straight backshell.

Connector mated pair # 2: connector plug with 90-degree backshell and connector receptacle with straight backshell.

Connector mated pair # 3: connector plug with straight backshell and connector receptacle with straight backshell.

9/ Tests are only applicable for connectors with nonmetallic inserts.

10/ Test one sample unit (connector, backshell, and dust cover) for each class designator (material and plating/finish designators A, blank) offered during the group C period.

4.5.4.14 Additional shell sizes. Retention of qualification for all shell sizes may be granted after completion of testing for one shell size as determined by the qualifying activity.

4.5.5 Qualification of additional connectors and backshells. Qualification of wall-mounted (panel-mounted) receptacles shall qualify straight- (in-line-) and straight jamnut-mounted receptacles. Each shell size shall be qualified separately.

4.5.5.1 Additional shell sizes. If a connector of one shell size is qualified, and connectors of a second shell size with similar design, construction, and materials meet the visual and mechanical, size, weight, identification marking, workmanship, screw thread, insert retention radial strength, insert retention axial strength, coupling engagement and disengagement torque, backshell and insert retention nut attachment, insertion loss, return loss, cable pull-out, external bending moment, cable seal flexing, mating durability, impact, crush, vibration, shock, water pressure, thermal shock, temperature cycling, and electromagnetic effects inspections herein, the connectors of the second shell size are qualified.

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4.5.5.2 Alternate insert materials. If a connector with an insert of one material is qualified, and connectors with an insert of a second material meet the interoperability, visual and mechanical, size, weight, identification marking, workmanship, insert retention radial strength, insert retention axial strength, terminus insertion and removal force, terminus retention force, maintenance aging, insertion loss, return loss, shock, temperature cycling, life aging, electromagnetic effects, flammability, fungus resistance, and ozone exposure inspections herein, the connectors with the second insert material are qualified. Connector interoperability shall be performed using previously qualified plug/receptacle connector with a minimum of two termini from all qualified vendors.

4.5.5.3 Alternate shell/backshell materials. If a connector with a shell/backshell of one material is qualified, and connectors with shells/backshells of a second material meet the visual and mechanical, size, weight, identification marking, workmanship, screw thread, insert retention radial strength, insert retention axial strength, coupling engagement and disengagement torque, backshell and insert retention nut attachment, insertion loss, return loss, cable pull out, external bending moment, mating durability, impact, crush, shock, water pressure, thermal shock, temperature humidity cycling, temperature cycling, life aging, sand and dust, electromagnetic effects, fluid immersion, salt spray, flammability, fungus resistance, and ozone exposure inspections herein, the connectors with shells/backshells of the second material are qualified.

4.5.5.4 45-degree and 90-degree backshells. If a straight backshell or a connector with a straight backshell is qualified for a shell size, and a 45- or 90-degree backshell, or connector with a 45- or 90-degree backshell, with similar design, construction, and materials meets the visual and mechanical, size, weight, identification marking, workmanship, screw threads, and backshell and insert retention nut attachment inspections herein, then the 45- or 90-degree backshell, or connector with a 45- or 90-degree backshell, are qualified for that shell size.

4.5.5.5 Joint qualification of connectors and termini. Connectors may be qualified using previously qualified termini or using unqualified termini. Connectors may be qualified using either multimode termini or single mode termini. Connectors may be qualified using both multimode termini and single mode termini. In all cases, the failure of any inspection by any terminus shall be cause for refusal to grant qualification approval.

4.5.5.6 Alternate plating/finish. If a connector, backshell, or dust cover of one finish/plating is qualified, and connectors, backshells, or dust covers of a second finish/plating meet visual and mechanical inspection, size, weight, identification marking, workmanship, backshell and insert retention nut attachment, connector coupling engagement and disengagement torque, interoperability, external bending moment, mating durability, impact, vibration, thermal shock, life aging, sand and dust, receptacle (conductive)-to-bulkhead conductivity (conductive finish only), electromagnetic effects, fluid immersion and salt spray inspections herein, the connectors, backshells or dust covers of the second plating shall be considered qualified.

4.6 Conformance inspection. Conformance inspection shall consist of the inspections and optical tests (see table VI) specified for group A inspection, group B inspection, and group C inspection. Inspections shall conform to applicable refinements and boundaries (see 6.3.5). Unless otherwise approved by the qualifying activity, additional connectors, backshells, and inserts qualified through reduced testing (see 4.5.5) shall be required to go through conformance tests.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection (see table VI).

4.6.1.1 Inspection lot. The inspection lot shall consist of the number of units offered for inspection at one time and all of the same design as covered by one specification sheet.

4.6.1.1.1 Sample unit. A sample unit shall be selected at random from the inspection lot. For conformance inspections, a sample unit shall consist of an individual unit of supply.

4.6.1.2 Group A inspection. Group A inspection shall consist of the inspection tests specified in table VI. All connector parts in the inspection sample shall be subjected to the inspection tests listed. Refer to 4.5.4.1 for the time intervals of Group A inspections for retention of qualification.

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4.6.1.2.1 Sampling plan. Group A inspection shall be performed by random sampling for product supplied under this specification. This random sampling shall be performed in accordance with MIL-STD-1678-1, Requirement 1304. The number of samples for random sampling inspected per lot size shall be as specified in the general inspection table of MIL-STD-1678-1, Requirement 1304. The required acceptance criterion shall be zero nonconformance regardless of sample size.

4.6.1.2.1.1 Group A sampling plan for occurrence of a nonconformance. If any nonconformance is found, 100 percent inspection shall be used until the qualifying activity is satisfied that the cause for the nonconformance has been corrected and will not reoccur. Until the nonconformance has been corrected, inspection with the requirement that data be provided (in lieu of inspection by attribute) may be specified by the qualifying activity.

4.6.1.2.2 Failures. One or more failures shall constitute group A inspection failure of the sample unit.

4.6.1.2.3 Disposition of sample units. Sample units that have failed any of the group A inspection tests shall not be shipped or submitted for group B testing.

4.6.2 Group B inspection. Group B inspection shall consist of the tests specified in table VI in the order shown. Group B inspection shall be performed on sample units selected from inspection lots which have passed group A inspection. Refer to 4.5.4.2 for the time intervals of Group B inspections for retention of qualification.

4.6.2.1 Sampling plan. A minimum of four sample units shall be selected from a lot of the same PIN within 24 months after the date of qualification and during every 24-month period thereafter, except when the total production in a 24-month period is less than 500 units of product or a total of 60 months have elapsed since the inspection was performed, in which case only two samples shall be tested. At no time shall the group B inspections be extended beyond 60 months. When a manufacturer is qualified for multiple shell sizes, the shell size selected for group B shall alternate as stated in table VII such that after the applicable number of consecutive group B inspections all qualified shell sizes have been inspected.

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TABLE VII. Component group B sampling plan by shell size.

Component	Group B Period 1	Group B Period 2	Group B Period 3	Group B Period 4
Connector 1/ 2/ 3/ 4/ 5/	11/MM	13/MM	15/SM	23/SM
Backshell 1/ 2/ 3/ 4/ 5/	11/MM	13/MM	15/SM	23/SM
Dust cover 1/ 2/	11	13	15	23
Integral connector with backshell 1/ 2/ 3/ 4/ 5/	11/MM	13/MM	15/SM	23/SM

NOTES:

1/ Shell size/fiber type such as "11/MM" where applicable shell sizes are 11, 13, 15 or 23 and where fiber types are SM = single mode, MM = multimode.

2/ If not qualified for all shall sizes in a particular row alternating inspections shall be performed for the qualified shell sizes. (Ex: If qualified for shall sizes 13 and 15, but not shell sizes 11 and 23, perform inspection on shell size 13 during Period 1, shell size 15 during Period 2, shell size 13 during Period 3, and shell size 15 during Period 4).

3/ Fabrication shall be performed in accordance with MIL-STD-1678-4, Requirement 4102.

4/ Applicable for the component configuration specified in each specification sheet unless a specification sheet is exempt due to qualification by similarity.

5/ Perform for component with PIN designators of:

Insert arrangement 1 (i.e., 2-, 4-, 8-, and 31-cavity positions for shell sizes 11, 13, 15, and 23, respectively).

Connector plug with pin terminus insert (style P), connector receptacle with socket terminus insert (style S).

Keying position number 1.

Cable outer diameter for cable specified in MIL-STD-1678-4, Requirement 4102.

Class (material with plating/finish). Use designator blank (aluminum with non-conductive finish) if more than one material with plating/finish is offered.

4.6.2.1.1 Sample unit preparation. Connectors shall be fully assembled into mated pairs using the types of cable specified in and fabricated in accordance with MIL-STD-1678-4, Requirement 4102. For mated connectors, full sealing capability shall be as specified (see 3.1).

4.6.2.1.1.1 Termini. The minimum number of active termini required for testing for each shell size shall be as follows:

a. A minimum of two termini shall be active in connectors of shell size 11.

b. A minimum of four termini shall be active in connectors of shell sizes 13 and 15.

c. A minimum of eight termini shall be active in connectors of shell size 23.

d. For shell sizes 15 and 23, active termini shall be placed at different locations in each of the four samples.

4.6.2.2 Failures. If one or more sample units fail to pass group B inspection, the lot from which the samples were selected shall be rejected.

4.6.2.3 Disposition of sample units. Sample units which have been subjected to group B inspection may be delivered on the contract or purchase order.

4.6.2.4 Rejected lots. Requirements regarding the rework of rejected lots shall be as identified in the qualification instructions (see 6.4.1).

4.6.3 Group C inspection. Group C inspection shall consist of the inspections specified in table VI, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed group A and B inspections. Refer to 4.5.6.3 for the time intervals of Group C inspections for retention of qualification.

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4.6.3.1 Sample unit preparation. Connectors shall be fully assembled into mated pairs using the types of cable specified in and fabricated in accordance with MIL-STD-1678-4, Requirement 4102. For mated connectors, full sealing capability shall be provided as specified (see 3.1).

4.6.3.1.1 Termini. The minimum number of active termini required for testing for each shell size shall be as follows: A minimum of two termini shall be active in connectors of shell size 11. A minimum of four termini shall be active in connectors of shell sizes 13 and 15. A minimum of eight termini shall be active in connectors of shell size 23. For shell sizes 15 and 23, the active termini shall be placed at different locations in each of the four samples.

4.6.3.2 Sampling plan. Group C inspection shall be performed on three connectors of the same PIN with their mating connectors 60 months after initial qualification and within each 5-year period thereafter. Connector mated pair shall be subjected to group I tests (see table VI). Connector mated pair from the group I tests shall be divided among the group II and III tests (see table VI). When a manufacturer is qualified for multiple shell sizes, then the shell size selected for group C shall alternate as stated in table VIII such that after the applicable number of consecutive group C inspections all qualified shell sizes have been inspected.

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TABLE VIII. Component group C sampling plan by shell size.

Component	Sampling plan (based on production volume)	Component as a part of a multiple fiber connector 1/	Fiber type 2/
Connector 3/, 4/, 5/	Select highest volume from 5-year period 6/	Connector Mated Pair 1 Connector Mated Pair 2 Connector Mated Pair 3	MM SM SM
Backshell 3/, 4/, 5/	Select highest volume from 5-year period 6/	Connector Mated Pair 1 Connector Mated Pair 2 Connector Mated Pair 3	MM SM SM
Integral connector with backshell 3/, 4/, 5/	Select highest volume from 5-year period 6/	Connector Mated Pair 1 Connector Mated Pair 2 Connector Mated Pair 3	MM SM SM

NOTES:

1/ See table VI for tests to be performed for each connector mated pair.

2/ Fiber type. SM = single mode, MM = multimode.

3/ Fabrication shall be performed in accordance with MIL-STD-1678-4, Requirement 4102.

4/ Applicable for the component configuration specified in each specification sheet unless a specification sheet is exempt due to qualification by similarity.

5/ Perform for component with PIN designators of:

Insert arrangement 1 (i.e., 2-, 4-, 8-, and 31-cavity positions for shell sizes 11, 13, 15, and 23, respectively).

Connector plug with pin terminus insert (style P), connector receptacle with socket terminus insert (style S). Keying position number 1.

Cable outer diameter for cable specified in MIL-STD-1678-4, Requirement 4102.

6/ The highest volume shell size from 5-year period shall be selected for each component specification sheet being offered. For connector specification sheets, this may encompass multiple specification sheets if the same connector (either connector plug or connector receptacle) is used for the connector (stand-alone) and for integral connectors with backshells. For backshell specification sheets, this may encompass three specification sheets if the same backshell design and parts are used for straight, 45-degree and 90-degree backshells.

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4.6.3.3 Failures. One or more specimen or sample unit failures shall constitute group C inspection failure.

4.6.3.4 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be shipped.

4.6.3.5 Noncompliance. Requirements regarding failure of group C inspection shall be as identified in the conformance inspection instructions (see 6.4.2).

4.7 Methods of inspection.

4.7.1 Equivalent test methods. The use of equivalent test methods shall be allowed provided the preparing activity and the qualifying activity have approved the use of that equivalent test method by that manufacturer (see 6.3.5).

4.7.2 Visual and mechanical inspection. Connector parts shall be examined in accordance with TIA-455-13 and in compliance with the data recording and evaluation criteria in Measurements 3101 through 3105 of MIL-STD-1678-3 to verify that materials, design, construction, and physical dimensions are in accordance with the applicable requirements (see 3.3, 3.4, 3.5, 3.6, 3.7, and 3.8).

4.7.2.1 Size. Each of the dimensions specified (see 3.1) for the connector parts shall be measured using calibrated measuring devices with the precision and accuracy appropriate for the tolerances specified (see 3.1). Dimensions shall be in accordance with the applicable specification sheets (see 3.10.1).

4.7.2.2 Weight. The connector parts shall be weighed using calibrated scales, having the range, precision, and accuracy appropriate for the tolerances specified (see 3.1 and 3.10.2).

4.7.2.3 Identification marking. Identification marking on connectors, backshells, and dust covers shall be visually examined and measured for conformance with the requirements of 3.10.3. Inspection for legible markings shall be performed from a distance of at least 6 inches with normal room lighting and without the aid of magnification.

4.7.2.4 Workmanship inspection. The connectors, backshells, and dust covers shall be visually examined to verify that they meet the workmanship requirements of 3.10.4.

4.7.2.5 Screw threads. Screw threads shall be checked after protective coating by means of ring and plug gauges (see MIL-STD-1373 for guidance).

4.7.3 Optical conformance test methods. The optical measurements shall be performed per the applicable TIA-455 series standard with the exception for adhering strictly to the setup and test procedure specified in the applicable measurement in Measurements 2101 through 2106 of MIL-STD-1678-2. Multiple fibers may not be concatenated during the measurement of change in optical transmittance or optical discontinuity. The center wavelength and light launch conditions for the specified fiber size and temperature range shall be as specified in MIL-STD-1678-2.

4.7.3.1 Insertion loss (see 3.11.1). The initial insertion loss of multimode connectors shall be measured in accordance with Measurement 2101 of MIL-STD-1678-2 using both an overfill launch (see 6.10.9) condition and a restricted launch condition in accordance with Measurement Support Process 2203 of MIL-STD-1678-2. Subsequent insertion loss tests shall be performed with the specified restricted launch condition in accordance with Measurement Support Process 2203 of MIL-STD-1678-2. The insertion loss of single mode connectors shall be measured in accordance with Measurement 2101 of MIL-STD-1678-2 using light launch conditions in accordance with Measurement Support Process 2203 of MIL-STD-1678-2.

4.7.3.2 Discontinuities (see 3.11.2). The connector shall be tested in accordance with Measurement 2104 of MIL-STD-1678-2 using test equipment having a time resolution sufficient to resolve discontinuities of duration not less than 50 microseconds (µs).

4.7.3.3 Crosstalk (see 3.11.3). Crosstalk shall be measured in accordance with Measurement 2103 of MIL-STD-1678-2.

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4.7.3.4 Change in optical transmittance (see 3.11.4). The change in optical transmittance shall be measured in accordance with Measurement 2102 of MIL-STD-1678-2 or by an equivalent method. The periodicity of the measurements shall be sufficient for the environmental test performed as approved by the qualifying activity. The use of a reference fiber to evaluate the change in optical transmittance due to exposure of the cable to the environmental tests is optional.

4.7.3.5 Return loss (see 3.11.5). Return loss shall be measured in accordance with Measurement 2105 of MIL-STD-1678-2 or by an equivalent method.

4.7.4 Functional inspections.

4.7.4.1 Insert retention radial strength (see 3.12.1). Unmated connector samples shall be tested for radial strength as follows. Counterpart test devices for plugs and receptacles shall be supplied by the connector manufacturer which are capable of applying radial torque forces between the insert and its shell body. Outermost termini positions or other means may be used for application of these torques, however, no damage shall occur to the insert as a result of the test exposure. A radial torque shall be applied up to the minimum specified (see 3.12.1) and held for 1 minute. The same torque shall then be applied in the opposite direction and held for 1 minute. Termini positions within the insert may be either empty or filled during the test.

4.7.4.2 Insert retention axial strength (see 3.12.2). Unmated connector samples shall be tested for axial strength as follows. An axial load shall be applied to either the front or rear face of the insert up to a minimum value of 100 pounds per square inch (0.69 Megapascal [MPa]) of insert face surface area. The load shall be held for a minimum of 1 minute and the axial displacement measured (see 3.12.2). The same pressure shall then be applied on the opposite face, held for 1 minute and the axial displacement measured. Termini positions within the insert may be either empty or filled during the test.

4.7.4.3 Terminus insertion and removal forces. Termini shall be inserted into an unmated connector using a terminus insertion tool and the force required to insert the terminus measured (see 3.12.3). A terminus removal tool shall then be engaged to unlock the terminus. The terminus shall be removed and the force required to remove the terminus measured. Socket terminus shall be used. When maintenance aging is performed, the terminus insertion and removal force tests performed at the first and last maintenance aging cycle can be used as the terminus insertion and removal force test; otherwise, it shall be performed in the sequence listed in tables V and VII.

4.7.4.4 Terminus retention force. Termini shall be inserted into an unmated connector and tested as follows. Termini shall be subjected to axial compressive loads applied to the front of the terminus tending to push the terminus to the rear of the connector insert. (Care must be exercised in the design of the force application mechanism so that it does not physically touch the optical fiber endface.) Axial loads shall be applied at a rate of 1.0 pound (4.4 Newtons) per second up to the maximum axial load specified (see 3.12.4). The maximum axial load specified in 3.12.4 shall be maintained for a minimum of 5 seconds.

4.7.4.1 Socket terminus. For a connector-only test, the socket terminus shall be tested in the connector with a pin insert cavity and with a socket insert cavity. For a combined connector and termini test, the socket terminus shall be tested in the connector with a socket insert cavity. A preload not greater than 3 pounds (13.4 Newtons) may be used to seat the terminus for the initial position measurement. The preload should be maximized for a socket (non-spring loaded) terminus. The axial displacement of the terminus from the pre-load position shall be measured while under the maximum axial load specified.

4.7.4.2 Pin terminus. For a combined connector and termini test, the pin terminus shall be tested in the connector with a pin insert cavity. A preload not greater than 3 pounds (13.4 Newtons) may be used to seat the terminus for the initial position measurement. The preload should be minimized a pin (spring loaded) terminus. The axial displacement of the terminus from the pre-load position shall be measured after the axial load is removed.

4.7.4.5 Maintenance aging (see 3.12.5). Unmated connectors shall be tested as follows. Termini shall be inserted and removed from the connector a minimum of 10 times. The termini selected for insertion and removal shall be the same termini that are monitored for optical performance. The force required to insert each terminus into the connector and the force required to remove each terminus from the connector shall be measured during the first and final maintenance aging cycle.

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4.7.4.6 Connector coupling engagement and disengagement torque (see 3.12.6). Unmated connector samples (or connector and dust cover samples) shall be tested as follows. The two mating parts shall be brought to a position where mechanical mating begins and the torque gauge is at zero indication. The parts shall then be fully mated or coupled and the torque required for mating shall be recorded. The torque gauge shall then be reset to zero indication. The mated parts shall be fully unmated and the torque required for unmating shall be recorded.

4.7.4.7 Backshell and insert retention nut attachment (see 3.12.7). Connector backshells and insert retention nuts shall be mated and unmated five times to their counterpart connectors. The torque specified in table III shall be applied to mate the backshell. The torque required to remove the backshell or insert retention nut shall be measured on the last unmating. The backshell or insert retention nut shall be visually examined in accordance with 4.7.2 after the test.

4.7.5 Mechanical test methods.

4.7.5.1 Cable pull out force (see 3.13.1). Mated connector samples shall be tested in accordance with TIA-455-6. The axial tensile load shall be applied up to the load specified and shall be maintained for 10 minutes. The change in optical transmittance shall be measured during and after the test (see 4.7.3.4). At the completion of the test, the connector shall be visually examined in accordance with 4.7.2.

4.7.5.2 External bending moment (see 3.13.2). Cable-connector assemblies shall be tested in accordance with the following procedure. The cabled receptacle shall be mounted as in normal service to a rigid wall (panel). After mating the plug and receptacle, a minimum bending moment of 300 inch-pounds (33.9 newton-meters) as measured from the wall (panel) shall be applied. The load (P) shall be applied across the smallest continuous exterior dimension of the plug connector backshell (see figure 2, note 4). This section may be circular or conical in the longitudinal direction. The load shall be applied at a rate of approximately 10 inch-pounds (1.1 newton-meters) per second until the required load is applied. The load shall be held for 1 minute. As an alternate to the load directly applied to the plug connector backshell, a fixture of any convenient design may be used as rigid bending moment test arm for the application of the applied load. Before mating the cabled plug to the receptacle, this rigid bending moment arm shall be secured to the rear of the plug shell. The fixture shall not provide support for the connector shell in front of the engaged threads. The change in optical transmittance shall be measured after the test (see 4.7.3.4).

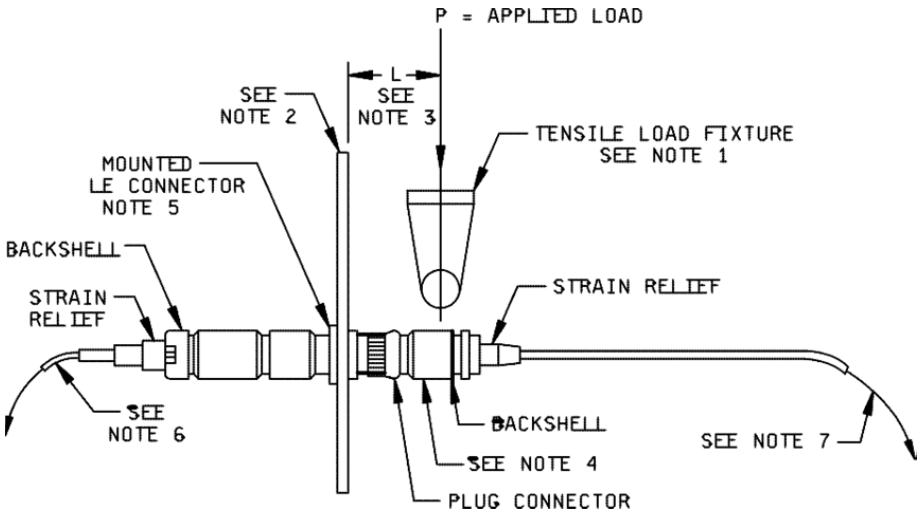


FIGURE 2. External bending moment test fixture and connector setup.

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NOTES:

1. Point of applied load P. Representation shows one means to apply the load P.
2. Test fixture 1. Test fixture simulating wall (rigid mounting panel) used to secure a receptacle connector in normal service.
3. Distance L is distance from plug connector side of test fixture to point of applied load.
4. Point of applied load. Load is applied on backshell at smallest, continuous section. The smallest, continuous section is defined as the smallest diameter on the backshell that is exposed and at least 2.5 inches (62.5 millimeters) in length. Section may circular or conical in longitudinal direction. If conical, load may be applied anywhere along longitudinal length. Alternate means to apply load by using a rigid bending moment arm is not shown.
5. Mount receptacle connector on side opposite in which receptacle connector will be mated to plug connector.
6. Fiber optic cable that connects to optical source end test instrumentation (optional).
7. Fiber optic cable that connects to detector end test instrumentation (optional).

FIGURE 2. External bending moment test fixture and connector setup – Continued.

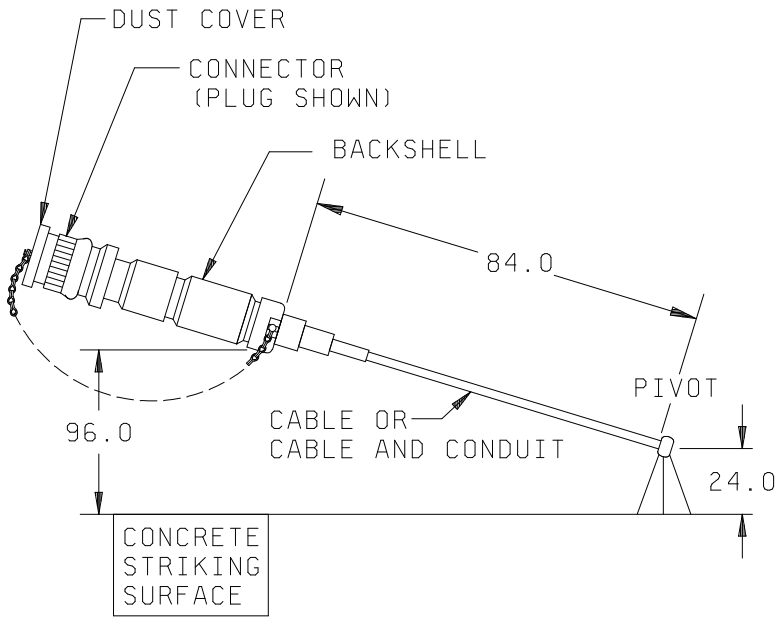
4.7.5.3 Cable seal flexing (see 3.13.3). Connector assemblies shall be tested in accordance with TIA/EIA-455-1. The connector assembly shall be exposed to 100 flexing cycles and the assembly rotated approximately 90 degrees in the flexing fixture and then exposed to another 100 flexing cycles. The cycling rate is not to exceed 14 cycles per minute if the test is performed manually. The connector assemblies shall be visually examined in accordance with 4.7.2 after the test.

4.7.5.4 Twist (see 3.13.4). Mated cable-connector assemblies shall be tested in accordance with TIA-455-36. The connector-held fixture shall be rotated 360 degrees at a rate of one cycle per 5 seconds for a total of 50 cycles. One cycle shall consist of a 360 degrees twist ±180 degrees about the neutral axis. The cable assemblies shall be stretched with minimum tension of 11 pounds (48.9 Newtons) to their maximum lengths and clamped at a distance of about 100 times the cable diameter from the connector to the tabletop. The connectors shall be visually examined in accordance with 4.7.2 after the test. The change in optical transmittance shall be measured during and after the test (see 4.7.3.4).

4.7.5.5 Mating durability (see 3.13.5). Connector plugs and receptacles shall be tested in accordance with TIA-455-21. Five hundred complete (plug and receptacle separating) cycles (mate and unmate) shall be accomplished by hand at a rate not to exceed 1 cycle per 15 seconds. The change in optical transmittance (see 4.7.3.4) shall be measured every 100 mating cycles during the test and after the test. Cleaning of the termini is permitted during and after completion of the test in order to meet the requirements of 3.11.4.

4.7.5.6 Impact (see 3.13.6). The unmated cable assemblies with backshell and dust cover shall be tested in accordance with method B of TIA/EIA-455-2 (see figure 3). Impacts shall be performed on the cable assembly with the plug connector. The dust cover shall be screwed onto the plug connector. The cable assembly shall be extended its full length from the test fixture. The plug shall be dropped eight times (the eight highest drops specified for the moderate service class in TIA/EIA-455-2) and rotated after each fall so that the connector strikes the impact pad in eight different radial positions. The test assemblies shall be visually examined for damage in accordance with 4.7.2, then mated. The connector termini may be cleaned after exposure and prior to mating. The change in optical transmittance shall be measured after the test (see 4.7.3.4).

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Inches	Millimeters
24	610
84	2,134
96	2,438

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Conduit or other means of stiffening the cable may be used to ensure that the impact occurs on the connector surface in the specified orientation.

FIGURE 3. Impact test fixture connector setup.

4.7.5.7 Crush (see 3.13.7). Connectors shall be tested in accordance with TIA/EIA-455-26, with the exception that the test sample shall be a connector. The test load shall be 281 pounds (1,250 Newtons), and the number of loading cycles shall be seven. The two load bearing surfaces may be faced with 1-inch thick rubber pads with a durometer between 60 and 75. Position the connector on the crush pads so that the plug, plug backshell, and portion of receptacle in front of the wall (panel) mounting is on the crush pads. The change in optical transmittance shall be measured during and after the test (see 4.7.3.4). The connector shall be visually examined in accordance with 4.7.2 after the test.

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4.7.6 Environmental test methods. Connectors with a composite connector part in accordance with 3.4, shall be unmated and re-mated after each environmental test. Change in optical transmittance measurements shall be performed prior to the connector being unmated and after the connector re-mating. Post-exposure optical transmittance measurements may be taken up to 24 hours after completion of the environmental exposure.

4.7.6.1 Thermal shock (see 3.14.2). Mated connectors shall be tested in accordance with Measurement 3304 of MIL-STD-1678-3. The temperature extremes shall be the specified non-operational temperature extremes. The change in optical transmittance shall be measured after the test (see 4.7.3.4). The connectors shall be visually examined in accordance with 4.7.2 after the test.

4.7.6.2 Temperature humidity cycling (see 3.14.3). Mated cable-connector assemblies shall be tested in accordance with Measurement 3302 of MIL-STD-1678-3. The subcycle shall be included in the testing. The change in optical transmittance shall be measured during and after the test (see 4.7.3.4).

4.7.6.3 Temperature cycling (see 3.14.4). Mated cable-connector assemblies shall be tested in accordance with Measurement 3301 of MIL-STD-1678-3. The change in optical transmittance shall be measured during and after the test (see 4.7.3.4). The connector assemblies shall be visually examined in accordance with 4.7.2 after the test.

4.7.6.4 Life aging. Mated connectors shall be tested in accordance with Measurement 3303 of MIL-STD-1678-3 and as specified (see 3.14.5). The specimens shall be exposed to dry air at between 230 and 239 °F (110 and 115 °C) for a period of 240 hours. The change in optical transmittance shall be monitored after the test in accordance with 4.7.3.4. The connectors shall be visually examined in accordance with 4.7.2 after the test. For connectors with dielectric inserts, the inserts shall be inspected for insert retention radial strength and insert retention axial strength after the test.

4.7.6.5 Freezing water immersion (see 3.14.6). Mated connector assemblies shall be tested in accordance with Measurement 3307 of MIL-STD-1678-3. The size of the water vessel shall be such that when the mated connectors are placed in the vessel, the mated connectors are within 150 millimeters of the sides and bottom of the vessel and within 150 millimeters of the surface of the water. The change in optical transmittance shall be monitored during and after the test in accordance with 4.7.3.4. At the completion of the test, the connectors shall be visually examined in accordance with 4.7.2. For the exposure at +14 °F (-10 °C), the water is considered completely frozen when the water temperature reading is less than +30.2 °F (-1 °C).

4.7.6.6 Sand and dust (see 3.14.7). Mated cable connector assemblies shall be tested in accordance with TIA-455-35 except as noted herein. Each connector shall be oriented in the chamber such that the coupling mechanism is in line with the oncoming air flow. The connectors shall be affixed in such a manner that the orientation of the connector does not change throughout the duration of the test. The 16-hour holding period of step 5.5 is not required. Step 5.6 may proceed immediately after reaching temperature stabilization. The change in optical transmittance shall be measured before the dust test, during the 6-hour exposure period of step 5.4, before step 5.6, during the 6 hour exposure period of step 5.6, and after the test. The connector coupling engagement and disengagement torque (see 4.7.4.6) shall be measured after the test. The connector shall be visually examined in accordance with 4.7.2, after cleaning, at the conclusion of the test.

4.7.6.7 Terminus cleaning (see 3.14.8). The optical face of the terminus shall be inspected and cleaned according to MIL-STD-2042-6, Method 6M1. The terminus shall not be removed from its operational position within the connector to facilitate cleaning.

4.7.6.8 Electromagnetic effects. Electromagnetic effects testing of receptacles (without backshells) mated to plugs with backshells and of receptacles (without backshells) mated to dust covers shall be conducted as follows. The test specimens (including terminated cable) shall be tested to determine the propagation characteristics (attenuation or conduction) of the specimen assembly. The shielding effectiveness of a shielded enclosure shall be verified in accordance with Measurement 3308 of MIL-STD-1678-3 at the discrete frequencies specified in table IX with the test specimen mounted in position. The dynamic range of the test setup shall be measured at each discrete frequency specified in table IX. The measured level of RF propagation through the test specimen installed in the shielded enclosure shall be determined in accordance with Measurement 3308 of MIL-STD-1678-3 at each discrete frequency specified in table IX.

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TABLE IX. Electromagnetic effects.

Frequency band	Frequencies (MHz)	Leakage attenuation (dB) minimum
VHF, UHF	100	58
	400	
	1,000	
Microwave	10,000	

4.7.6.8.1 Antenna placement. Antenna type and placement shall be in accordance with Measurement 3308 of MIL-STD-1673, with the exception that antenna types shall be in accordance with table X. For planewave measurements in the perpendicular direction, the transmit antenna shall be 1 meter above the test specimen cable.

TABLE X. Antenna types.

Frequency Band	Test frequencies
VHF, UHF	Log periodic or dipole
Microwave	Horn

4.7.6.9 Fluid immersion (see 3.14.10). Connector assemblies and separate polymeric samples shall be exposed to each of the fluids identified in accordance with Measurement 3409 of MIL-STD-1678-3 for test temperature range 1. One connector assembly shall be prepared for each fluid. The connector assembly shall consist of a connector dust cover (plug or receptacle), connector (plug or receptacle), backshell, and from 0.3 to 1 meter (12 to 39 inches) of multiple fiber cable exiting the backshell. Multiple fiber cable end inside the backshell shall be strain relieved with backshell seals in place. Multiple fiber cable end outside the backshell shall be adequately sealed. Polymeric samples shall consist of polymeric parts from one connector mated pair (which includes the backshell). Sample preconditioning shall be done under ambient conditions for a minimum of 4 hours. Each connector assembly shall be completely immersed in one fluid except for a short length (approximately 6 inches [152 millimeters] or less) at the exposed cable end. Each connector assembly and sample of each polymeric material shall be completely dried after each immersion. After testing, the connector assembly shall be visually examined for fluid penetration into the shell body and the connector junction region. Visual inspection for loss of sealing capacity (fluid penetration) shall be done by removing dust cover. Partial disassembly, opening of backshell at rear most nut on backshell, not backshell to connector coupling nut, of each connector assembly shall be performed later in the visual examination as one means to determine if there is any loss of sealing capacity (fluid penetration).

4.7.6.10 Salt spray (corrosion) (see 3.14.11). Mated cable-connector assemblies shall be tested in accordance with Measurement 3402 of MIL-STD-1678-3. The exposure time shall be 500 hours, and the exposure temperature shall be 95 °F (35 °C). After test exposure, the assemblies shall be externally cleaned and examined under three-power magnification for salt penetration into the connector junction area and damage to external parts.

4.7.6.11 Flammability (see 3.14.12). Mated and unmated cable-connector assemblies shall be tested in accordance with EIA-364-81 and as specified herein. Mated assemblies shall be exposed to a 0.75-inch (19-millimeter) flame height applied for 10 seconds to the region of the mated pair interface. The change in optical transmittance shall be measured during the test, and after the test once the test sample has returned to room temperature (see 4.7.3.4). The sample shall then be demated, and the unmated connector assembly with backshell and dust cover shall be exposed to a 1.50-inch (38.1-millimeter) flame height applied for 60 seconds to the backshell-cable interface region. For connectors with dielectric inserts, the connector samples shall be exposed to a third flame with a 1.50-inch (38.1-millimeter) flame height. The connector samples shall be remated and the flame shall be applied for 60 seconds to the region of the mated pair interface.

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4.7.6.12 Fungus resistance (see 3.14.13). Polymeric connector materials not listed as fungus inert in Guideline 4 of MIL-HDBK-454 shall be tested in accordance with Measurement 3401 of MIL-STD-1678-3 for a duration of 28 days.

4.7.6.13 Ozone exposure (see 3.14.14). Polymeric connector parts shall be tested to Measurement 3404 of MIL-STD-1678-3 and exposed to an ozone concentration of 100 to 150 parts per million at a temperature of 158 to 167 °F (70 to 75 °C) and an air velocity not less than 0.6 meters per second (m/s) for 2 hours. The ozone test apparatus and ozone measuring device shall be in accordance with ASTM D1149. If a polymeric part is expanded on the connector, the part shall be tested at the same level of expansion.

4.7.6.14 Vibration (see 3.14.15). Mated cable-connector assemblies shall be tested in accordance with Measurement 3201 of MIL-STD-1678-3 (test condition B for multiple termini connectors). Optical discontinuities shall be measured during each test (see 4.7.3.5). For connectors of shell sizes 13, 15, and 23, a minimum of four termini shall be monitored for discontinuity. The change in optical transmittance shall be measured after the test (see 4.7.3.4). The connector shall be visually examined in accordance with 4.7.2 after the test. Retightening of the connector after each axis and test condition is permitted.

4.7.6.15 Shock (see 3.14.16). Mated cable-connector assemblies shall be tested in accordance with MIL-STD-901, grade A, class I and Measurement 3202 of MIL-STD-1678-3. Optical discontinuities shall be measured during the test (see 4.7.3.2). For connectors of shell sizes 13, 15, and 23, a minimum of four termini shall be monitored for discontinuity. The change in optical transmittance shall be measured after the test (see 4.7.3.4). The connector shall be visually examined in accordance with 4.7.2 after the test. Retightening of the connector after each blow is permitted. Standard shock fixture 4A for bulkhead mounting shall be used.

4.7.6.16 Water pressure (see 3.14.17). Mated cable-connector assemblies shall be tested for water pressure susceptibility as follows. The assemblies shall be immersed in fresh water to an equivalent depth of 32 feet (9.8 meters) for a period of 48 hours. The water temperature shall be maintained between 50 and 95 °F (10 and 35 °C) during the exposure period. The connector assemblies shall be externally cleaned and unmated, and the backshell shall be removed (except for connectors with integral backshells) and visually examined for water penetration into the connector. The change in optical transmittance shall be measured after the test (see 4.7.3.4). Accessing the connector via backshell coupling nut for visual examination is permitted.

4.7.6.17 Receptacle (conductive)-to-bulkhead conductivity (see 3.14.18). The voltage drop across the conductive receptacle mounted to the bulkhead shall be tested in accordance with test procedure EIA-364-83 when measured from a point on the front receptacle surface, the forward most surface, to a point on the bulkhead next to, but not in contact with, the mounting flange. The probes shall not puncture or otherwise damage the connector finish.

4.8 Interoperability (see 3.4.3). Connectors and backshells shall be tested as specified in 4.8.1 and 4.8.2.

4.8.1 Connector interoperability. Unless otherwise specified (see 3.1), qualified termini shall be inserted into and shall fully populate connector specimens as specified in table XI. Insertion loss shall be measured in accordance with 4.7.3.1. The terminus insertion and removal forces shall be measured in accordance with 4.7.4.3. The terminus retention force shall be measured in accordance with 4.7.4.4.

TABLE XI. Connector interoperability test configurations.

Configuration no.	Connector receptacle	Connector plug
1	Qualified	Candidate
2	Candidate	Qualified
3	Candidate	Candidate

4.8.2 Backshell interoperability. Unless otherwise specified (see 3.1), qualified termini shall be inserted into and shall fully populate qualified connector plugs and receptacles with backshells as specified in table XII. Insertion loss shall be measured in accordance with 4.7.3.1.

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TABLE XII. Connector interoperability test configurations.

Configuration No.	Receptacle backshell	Plug backshell
1	Candidate	Candidate

4.9 Toxicity. When specified (see 3.1), a Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.3.5 as required by the qualifying activity. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the materials using data provided by the manufacturer/distributor to the NMCCPHC (see 3.3.5 and 6.6).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fiber optic connectors covered by this specification are intended for use in military applications where their performance characteristics are required. The connectors covered by this specification are unique due to the fact that these items must be able to operate satisfactorily in systems under the following demanding conditions: 10 g's vibration, over 1,000 g's of shock, temperature excursions from -40 to +158 °F (-40 to +70 °C), and mechanically harsh conditions. For applications with vibration exposures greater than 75 Hz, lock wire and connector coupling torques higher than those specified in table III should be considered. A gasket is not provided with receptacles, and a gasket may be required to achieve shielding effectiveness specified herein. For applications using dummy termini, lock wire and connector coupling torques higher than those specified in table III should be considered. In addition, these requirements are verified under a qualification system. Commercial connectors are not designed to withstand these environmental conditions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. PIN (see 1.3).

c. Quantity of connectors required.

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 28876 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160, CommandStandards@navy.mil; however, information pertaining to qualification of products may be obtained from the agent for the qualifying activity: DLA Land and Maritime; VQP, P.O. Box 3990, Columbus, OH 43218-3990. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

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6.3.1 Conformity to qualification sample. It is understood that connectors supplied under the contract should be identical in every respect to the qualification sample tested and found satisfactory, except for changes previously approved by the Government. Any unapproved changes from the qualification sample should constitute cause for rejection.

6.3.2 Provisions governing qualification. Copies of SD-6, "Provisions Governing Qualification" may be obtained using the ASSIST Online database at https://quicksearch.dla.mil/.

6.3.3 Verification program. A verification program is to be established and maintained in accordance with MIL-STD-790 or equivalent standard. Evidence of such compliance will be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification. The verification system procedures, planning, and all other documentation and data that comprise the verification system is to be available to the Government for review. The Government may perform any necessary inspections, verifications, and evaluations to ascertain conformance to the requirements and adequacy of the implementing procedures (see 4.3).

6.3.4 Test equipment and inspection facilities. Provision for test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspections is to be the responsibility of the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment is to be in accordance with NCSL-Z540-3.

6.3.5 Inspection refinements and boundaries. MIL-STD-1678-2, MIL-STD-1678-3, or MIL-STD-1678-5, as applicable, provide further details to refine or bound (restrict) the performance (including setup and processes) of each optical, mechanical, environmental, and material test and inspection

6.3.6 Optical transmittance instrumentation stability. Optical transmittance instrumentation should be subjected to the following stability tests before qualification testing is performed. The first test should consist of measuring the transmitted power through each channel once every minute for a 4-hour period. The second test should consist of measuring the transmitted power through each channel once every 30 minutes for a 96-hour period. The data for each channel should be analyzed to determine average transmittance, minimum and maximum transmittance, the standard deviation of the transmittance, and the minimum and maximum percent deviation of the transmittance.

6.3.7 Government witnessing of qualification tests. The qualification activity may require that a government witness be present during the following tests: size, screw threads, connector assembly (test sample) fabrication, initial insertion loss, terminus insertion and removal forces, cable pull out force, mating durability, temperature cycling, mechanical shock, salt spray, and thermal shock.

6.3.8 Electromagnetic effects test documentation. Electromagnetic effects test documentation should include a detailed test plan and test report. The test plan should detail the specific test setups and procedures, the test facility, and the test profiles that will be used. The test report should detail the exact procedures followed, the equipment used for each test phase, calibration dates of all test equipment, test results in graphical and tabular format, photographs/sketches of the test setups, the test data sheets, and an indication of pass/fail. The test data sheets should include test sample identification, ambient temperature and humidity values, dates and times of the test initiation and completion, names and initials of the test personnel, and data tables. The data tables should include the frequency at which each measurement was conducted, the field propagation/orientation, enclosure shielding effectiveness, reference level of the transmitter, dynamic range of the measurement, measured level for the test, the propagated RF attenuation, the required RF attenuation, and an indication of pass/fail. The reference level should include the source output, receiver sensitivity, and antenna gain.

6.4 Conformance inspection.

6.4.1 Rejected lots. If a group B inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units (if applicable), and resubmit the lot for inspection.

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6.4.2 Noncompliance. If a sample fails to pass group C inspections, the manufacturer should notify the qualifying activity of the failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, and so forth, and which are considered subject to the same failure. Acceptance of the product should be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection should be repeated on additional sample units (all inspection tests or the inspection test which the original sample failed, at the option of the Government). Group A and B inspection may be reinstituted; however, final acceptance should be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken should be furnished to the cognizant inspection activity and the qualifying activity.

6.5 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of dating of this document, the Environmental Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals is available on their website at https://archive.epa.gov/epawaste/hazard/wastemin/web/html/priority.html. Included in the EPA list of 31 priority chemicals are cadmium, lead, mercury, and hexavalent chromium. Use of the materials on the list should be minimized or eliminated unless needed to meet the requirements specified herein (see section 3).

6.6 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, https://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx."

6.7 Change effectivity. Unless otherwise specified by the preparing activity or qualifying activity, all changes from MIL-PRF-28876E are to become effective on the date of publication of MIL-PRF-28876F. Production and manufacturing is to be in accordance with MIL-PRF-28876F. Manufacturers QPL-listed to MIL-PRF-28876E that have concerns regarding possible changes to retention reporting requirements should contact the qualifying activity for clarification.

6.8 Disposition of Stock. Disposition of stock is provided under the following:

a. A period of 90 days is provided for disposition of stock as described herein. This period of 90 days does not include the manufacturing of new parts to MIL-PRF-28876E; it is only to allow assembly and marking of parts produced prior to publication of MIL-PRF-28876F.

b. Manufacturers, selling agents, or distributors are permitted to assemble, mark, and date-code parts during the 90-day period following previously qualified product processes. During the 90-day period, connectors or accessories produced in accordance with MIL-PRF-28876E, where parts are not in accordance with MIL-PRF-28876F, are not to be mixed with parts produced in accordance with MIL-PRF-28876E within a single QPL part (i.e., delivered product is not permitted to contain parts produced to both MIL-PRF-28876E and MIL-PRF-28876F, unless all parts meet the requirements of MIL-PRF-28876F).

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Disposition of Stock

Activity	Timeline			
	MIL-PRF-28876E	**MIL-PRF-28887F publication date**	**MIL-PRF-28876F publication date + ≤90 days**	**MIL-PRF-28876F publication date + >90days**
Production and manufacturing of components to MIL-PRF-28876E	Yes	Stops	No	No
Assemble and date using MIL-PRF-28876E parts	Yes	Yes [1/]	Yes [1/]	Stops
Ship MIL-PRF-28876E parts	Yes	Yes [1/]	Yes [1/]	Yes [1/]
Production and manufacturing of components to MIL-PRF-28876F	N/A	Starts [2/] [3/]	Yes [3/]	Yes [3/]
Assemble, date, and ship using MIL-PRF-28876F parts	N/A	Yes [3/]	Yes [3/]	Yes [3/]
Assembly/mix of previous and current revision parts [4/]	N/A	Not permitted	Not permitted	Not permitted

NOTES:

[1/] Product produced to MIL-PRF-28876E is to have been manufactured, assembled, and dated during timeline allowed herein (see row 1 and row 2 of this table). No additional fabrication or plating of lower level components is permitted after the MIL-PRF-28876F publication date. Manufacturers, selling agents, or distributors shipping product produced to MIL-PRF-28876E (e.g., cadmium parts) are to ensure product produced to MIL-PRF-28876E (e.g., cadmium parts) is acceptable to procuring activity.

[2/] Production and manufacturing of components to MIL-PRF-28876F prior to its publication date may be performed at the risk of the manufacturer. Where a manufacturer chooses to start such production prior to publication of MIL-PRF-28876F, the manufacturer has an obligation to deliver items meeting all MIL-PRF-28876F requirements and production of components are to be identical in every respect to the qualification sample tested. The manufacturer cannot certify, mark, ship, or advertise they are qualified until a notification of qualification letter has been issued. This clarification does not supersede any qualification or conformance requirements.

[3/] Includes “HA” marking.

[4/] No mixing of parts from MIL-PRF-28876E and MIL-PRF-28876F where MIL-PRF-28876E parts are not in accordance with MIL-PRF-28876F.

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6.9 Cadmium alternative transition. Since the early 1990s, the U.S. Navy has expressed interest in moving away from the use of cadmium (along with hexavalent chromium) as a protective finish for connectors and their accessories. In an effort to replace cadmium and the hexavalent chromium topcoat as a connector and accessory finish for the new ship designs, the U.S. Navy established a Cadmium Alternative Working Group comprised of government and industry to evaluate cadmium-alternative finishes. Changes with regard to the cadmium-alternative finishes in this revision include the addition of Hard Anodized with PTFE finish (see 3.3.2.3) and the addition of Tin Zinc Finish (see 3.3.2.4). Manufacturers on the Qualified Parts List (QPL) are to transition product and production in accordance with the disposition of stock (see 6.8). Parts previously provided in accordance with MIL-PRF-28876 E will be provided with the HA-PTFE finish (see 6.8). During and after transition (see 6.8) M28876 QPL manufacturers, procuring activities, and end users are to use platings of like finish in cable assembly components (dust cover, connector, and backshell).

6.10 Definitions. Definitions and terms are in accordance with EIA-440 and as stated below.

6.10.1 70/70 restricted launch. A 70/70 restricted launch is beam optics launch with a 70 percent spot size and source aperture equal to 70 percent of the fiber numerical aperture.

6.10.2 Backshell. The backshell attaches to the rear of the connector shell, provides for environmental sealing of the connector, and provides for cable strain relief. The backshell also holds the connector insert in position within the connector shell, using an integral or separate insert retaining sleeve. The backshell includes the seal or sealing mechanism which seals to a plug or receptacle connector.

6.10.3 Connector. The connector is the entire cable termination assembly and is composed of the connector shell, connector insert, and backshell or insert retention nut.

6.10.4 Dust cover. The dust cover attaches to the front of the connector or receptacle and provides environmental sealing of the connector or receptacle when the connector or receptacle are in the unmated condition.

6.10.5 Insert. The insert is the interior portion of the connector which holds and aligns the optical termini.

6.10.6 Insert retaining sleeve. The insert retaining sleeve holds the insert in place and can be integral or separate to retention nut or backshell.

6.10.7 Insert retention nut. The insert retention nut attaches to the rear of the connector shell and holds the connector insert and insert retaining sleeve in position in the connector shell when the connector is used without a backshell. The insert retention nut utilizes an integral or a separate insert retaining sleeve.

6.10.8 Insertion loss. Insertion loss is the radiant power loss (dB) caused by absorption, scattering, diffusion, leaky waves, dispersion, microbends, macrobends, reflection, radiation, or other causes when a connector is inserted into the system.

6.10.9 Overfill launch. An overfill launch is a launch with a source spot size at least 100 percent the fiber spot size and source aperture at least 100 percent of the fiber numerical aperture.

6.10.10 Protective covering. A protective covering is a disposable protective cap or cover.

6.10.11 Shell. The shell is the front portion of the connector which holds the connector insert and contains the connector coupling mechanism. Shells are either of the plug or receptacle configuration.

6.10.12 Terminus. Terminus is the part of the connector that provides a means of positioning and holding the fiber within the connector.

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6.11 Subject term (key word) listing.

Backshells

Cable, fiber optic

Covers, protective

Dust covers

Environmental resistant

Epoxies

Inserts

Optical performance requirements

Screw threads

Strain relief, cable

Test plugs

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONNECTOR INTERFACE DIMENSIONS

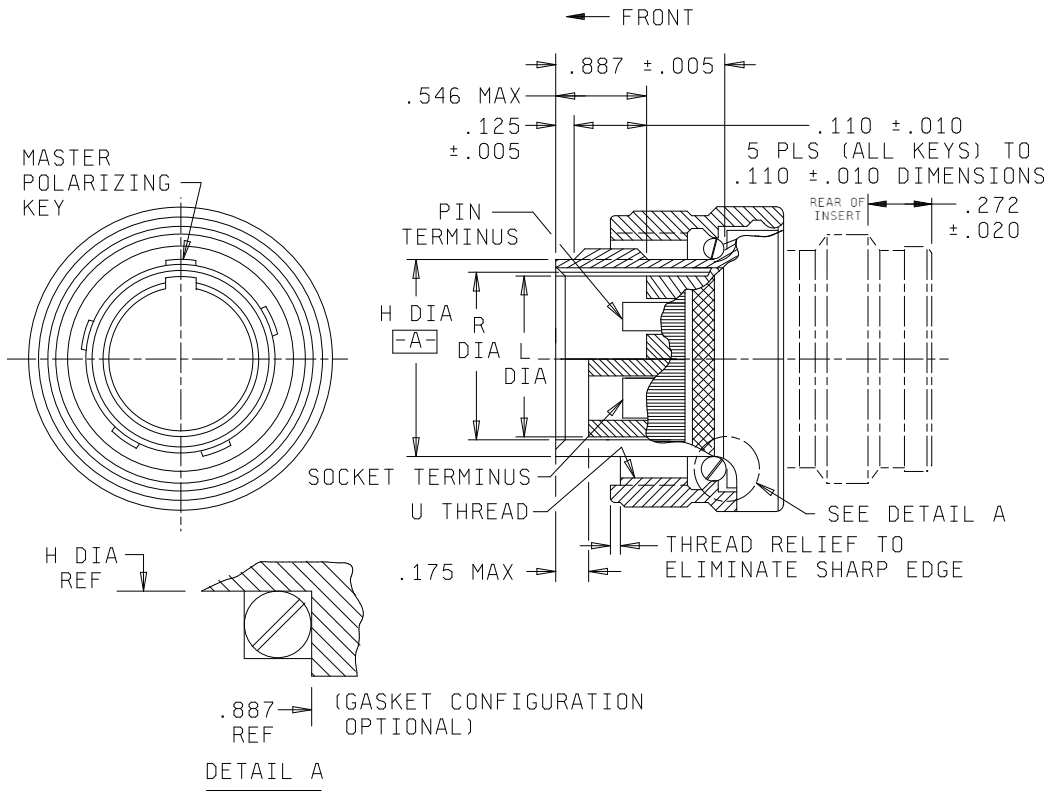
A.1 SCOPE

A.1.1 Scope. This appendix lists the connector interface dimensions. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 DRAWINGS

A.2.1 Interface dimension. The connector interface dimension drawings are listed on figures A-1 through A-8.

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Inches	Millimeters
0.005	0.13
0.010	0.25
0.020	0.51
0.110	2.79
0.125	3.18
0.175	4.45
0.272	6.91
0.546	13.87
0.887	22.53

FIGURE A-1. Interface dimensions, connector, plug, fiber optic.

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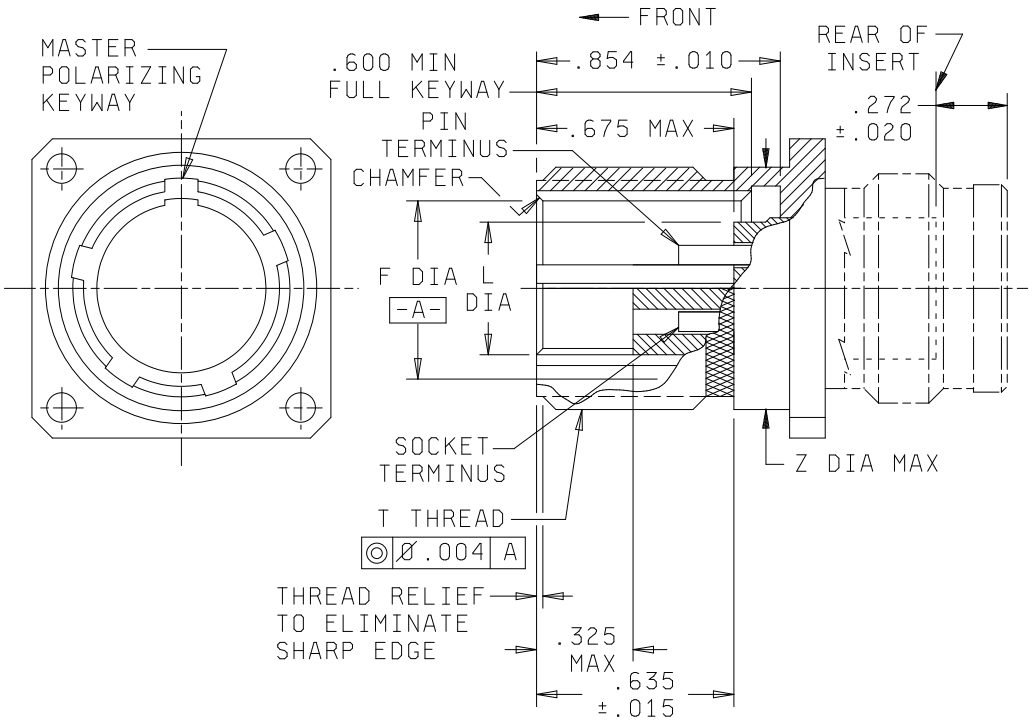
Shell size	U thread, class 2B	H dia	R shell I.D.	L insert dia
11	0.750-0.1P- 0.2L-D.S.	0.502 (12.75) 0.492 (12.50)	0.383 (9.73) 0.373 (9.47)	0.365 (9.27) 0.361 (9.17)
13	875-0.1P- 0.2L-D.S.	0.626 (15.90) 0.616 (15.65)	0.505 (12.83) 0.495 (12.57)	0.488 (12.40) 0.483 (12.27)
15	1.062-0.1P- 0.2L-D.S.	0.798 (20.27) 0.788 (20.02)	0.683 (17.35) 0.673 (17.09)	0.666 (16.92) 0.661 (16.79)
23	1.500-0.1P- 0.2L-D.S.	1.220 (30.99) 1.210 (30.73)	1.089 (27.66) 1.079 (27.41)	1.073 (27.25) 1.063 (27.00)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Dimensions apply to plated/finished part.
5. Mating key positions and dimensions are shown on figure A-3.
6. This design information establishes connector intermating criteria.
7. Rear end connector design for attachment of non-rotatable backshell is shown on figure A-6.
8. See MIL-PRF-28876/6 through MIL-PRF-28876/9 for appropriate plug outer configuration dimensions.
9. See figures A-7 and A-8 for backshell and insert retention nut mating dimensions.

FIGURE A-1. Interface dimensions, connector, plug, fiber optic – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.



Inch	Millimeters
0.004	0.10
0.010	0.25
0.015	0.38
0.20	0.51
0.272	6.91
0.325	8.26
0.600	15.24
0.635	16.13
0.675	17.14
0.854	21.69

FIGURE A-2. Interface dimensions, connector, receptacle, fiber optic.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.

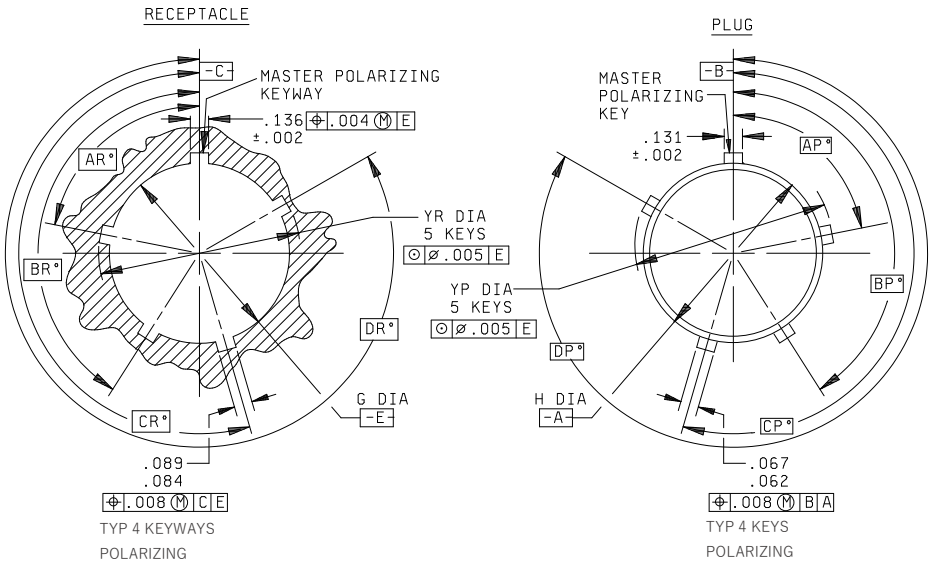
Shell size	T thread, class 2A	F shell I.D.	L insert dia	Z dia (max)
11	0.750-0.1P-	0.515 (13.08)	0.365 (9.27)	0.750 (19.05)
	0.2L-D.S.	0.506 (12.85)	0.361 (9.17)	
13	875-0.1P-	0.639 (16.23)	0.488 (12.40)	0.875 (22.23)
	0.2L-D.S.	0.630 (16.00)	0.483 (12.27)	
15	1.062-0.1P-	0.809 (20.55)	0.666 (16.92)	1.062 (26.97)
	0.2L-D.S.	0.800 (20.32)	0.661 (16.79)	
23	1.500-0.1P-	1.233 (31.32)	1.073 (27.25)	1.500 (38.10)
	0.2L-D.S.	1.224 (31.09)	1.063 (27.00)	

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Dimensions apply to plated/finished part.
5. Mating key positions and dimensions are shown on figure A-3.
6. This design information establishes connector intermating criteria.
7. Rear end connector design for attachment of non-rotatable backshell is shown on figure A-6.
8. See MIL-PRF-28876/1 through MIL-PRF-28876/5 and MIL-PRF-28876/11 through MIL-PRF-28876/14 for appropriate receptacle outer configuration dimensions.
9. See figures A-7 and A-8 for backshell and insert retention nut mating dimensions.

FIGURE A-2. Interface dimensions, connector, receptacle, fiber optic – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.



Inch	Millimeters
0.002	0.05
0.004	0.10
0.005	0.13
0.008	0.20
0.062	1.58
0.067	1.70
0.084	2.13
0.089	2.26
0.131	3.33
0.136	3.45

Shell size	G dia	H dia
11	0.515 (13.08)	0.502 (12.75)
	0.505 (12.83)	0.492 (12.50)
13	0.639 (16.23)	0.626 (15.90)
	0.629 (15.98)	0.616 (15.65)
15	0.809 (20.55)	0.798 (20.27)
	0.800 (20.32)	0.788 (20.02)
23	1.233 (31.32)	1.220 (30.99)
	1.223 (31.06)	1.210 (30.73)

FIGURE A-3. Connector, fiber optic, position of key and keyway mating.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.

Shell size	Key and keyway arrangement	AR° or AP° BSC [1/]	BR° or BP° BSC	CR° or CP° BSC	DR° or DP° BSC	YP dia	YR dia
11	1	95	141	208	236	0.559 (14.20)	0.581 (14.76)
	2	113	156	182	292		
	3	90	145	195	252		
	4	53	156	220	255	0.551 (14.00)	0.569 (14.45)
	5	119	146	176	298		
	6	51	141	184	242		
13	1	95	141	208	236	0.683 (17.35)	0.705 (17.91)
	2	113	156	182	292		
	3	90	145	195	252		
	4	53	156	220	255	0.675 (17.14)	0.693 (17.60)
	5	119	146	176	298		
	6	51	141	184	242		
15	1	80	142	196	293	0.855 (21.72)	0.877 (22.28)
	2	135	170	200	310		
	3	49	169	200	244		
	4	66	140	200	257	0.847 (21.51)	0.865 (21.97)
	5	62	145	180	280		
	6	79	153	197	272		
23	1	80	142	196	293	1.276 (32.41)	1.301 (33.04)
	2	135	170	200	310		
	3	49	169	200	244		
	4	66	140	200	257	1.267 (32.18)	1.288 (32.71)
	5	62	145	180	280		
	6	79	153	197	272		

FOOTNOTE:

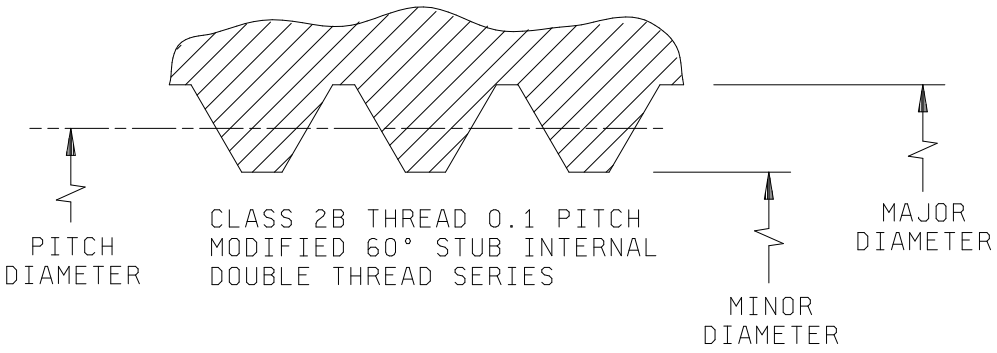
[1/] Basic (BSC).

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Dimensions apply to plated/finished part.

FIGURE A-3. Connector, fiber optic, position of key and keyway mating – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.



INTERNAL THREADS

Inch	Millimeters	Inches	Millimeters
0.100	2.54	0.8950	22.733
0.200	5.10	1.0025	25.464
0.7042	17.877	1.0145	25.768
0.7142	18.141	1.0285	26.124
0.7240	18.390	1.0405	26.429
0.7340	18.644	1.0625	26.988
0.7500	19.050	1.0665	27.089
0.7540	19.152	1.0865	27.597
0.7700	19.588	1.4400	36.576
0.8292	21.062	1.4520	36.881
0.8392	21.316	1.4660	37.236
0.8490	21.565	1.4780	37.541
0.8590	21.819	1.5000	38.100
0.8750	22.225	1.5040	38.202
0.8790	22.327	1.5240	38.710

FIGURE A-4. Connector mating threads (internal).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.

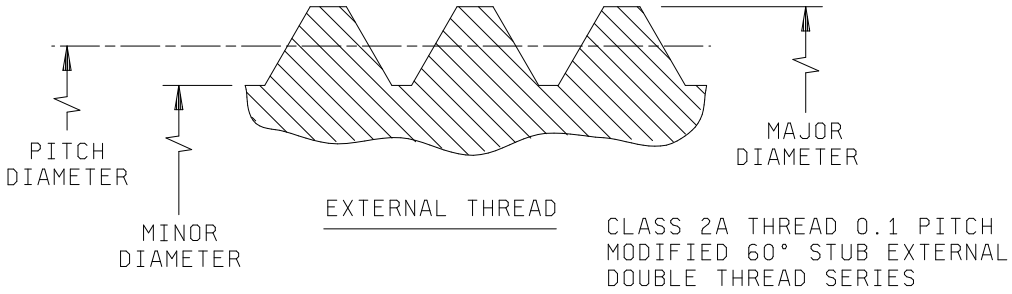
Shell size	Designation		
	Thread size	Pitch	Lead
11	0.7500 (19.05)	0.1 (2.54)	0.2 (5.08)
13	0.8750 (22.22)	0.1 (2.54)	0.2 (5.08)
15	1.0625 (26.99)	0.1 (2.54)	0.2 (5.08)
23	1.5000 (38.10)	0.1 (2.54)	0.2 (5.08)

Shell size	Internal thread limits of size					
	Minor diameter		Pitch diameter		Major diameter	
	Limits		Limits		Limits	
	Max	Min	Max	Min	Max	Min
11	0.7142 (18.14)	0.7042 (17.89)	0.7340 (18.64)	0.7240 (18.39)	0.7700 (19.56)	0.7540 (19.15)
13	0.8392 (21.31)	0.8292 (21.06)	0.8590 (21.82)	0.8490 (21.56)	0.8950 (22.73)	0.8790 (22.33)
15	1.0145 (25.77)	1.0025 (25.46)	1.0405 (26.43)	1.0285 (26.12)	1.0865 (27.60)	1.0665 (27.09)
23	1.4520 (36.88)	1.4400 (36.58)	1.4780 (37.54)	1.4660 (37.24)	1.5240 (38.71)	1.5040 (38.20)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE A-4. Connector mating threads (internal) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.



Inch	Millimeters	Inches	Millimeters
0.100	2.54	0.8750	22.225
0.200	5.10	0.9705	24.651
0.6785	17.234	0.9885	25.108
0.6925	17.590	1.0165	25.819
0.7145	18.148	1.0265	26.073
0.7225	18.352	1.0485	26.632
0.7405	18.809	1.0605	26.937
0.7485	19.012	1.0625	26.988
0.7500	19.050	1.4080	35.763
0.8035	20.409	1.4260	36.220
0.8175	20.764	1.4540	36.932
0.8395	21.323	1.4640	37.186
0.8475	21.526	1.4860	37.744
0.8655	21.984	1.4980	38.049
0.8735	22.187	1.5000	38.100

Shell size	Designation		
	Thread size	Pitch	Lead
11	0.7500 (19.05)	0.1 (2.54)	0.2 (5.08)
13	0.8750 (22.23)	0.1 (2.54)	0.2 (5.08)
15	1.0625 (26.99)	0.1 (2.54)	0.2 (5.08)
23	1.5000 (38.10)	0.1 (2.54)	0.2 (5.08)

FIGURE A-5. Connector mating threads (external).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.

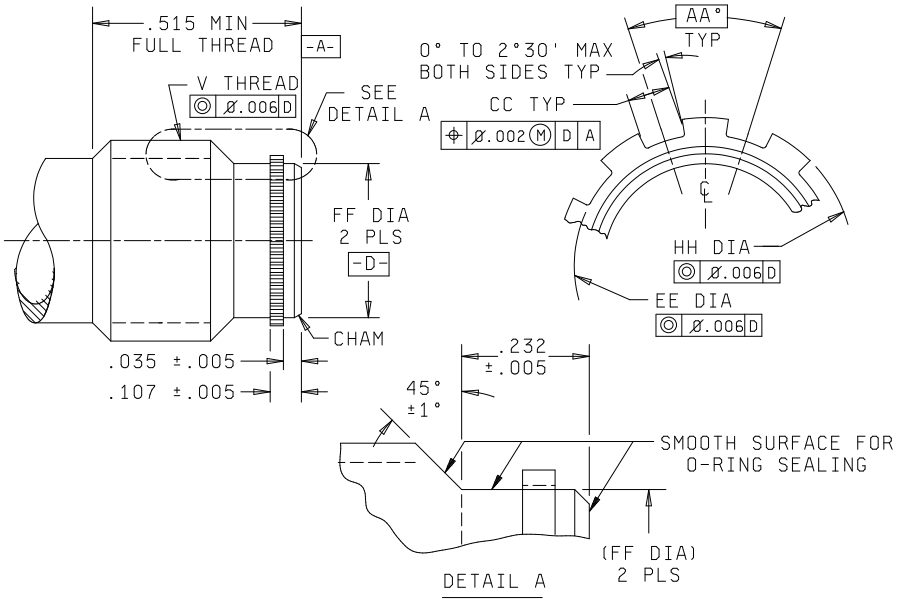
Shell size	Internal thread limits of size					
	Minor diameter		Pitch diameter		Major diameter	
	Limits		Limits		Limits	
	Max	Min	Max	Min	Max	Min
11	0.7485 (19.01)	0.7405 (18.81)	0.7225 (18.35)	0.7145 (18.15)	0.6925 (17.59)	0.6785 (17.23)
13	0.8735 (22.19)	0.8655 (21.98)	0.8475 (21.53)	0.8395 (21.32)	0.8175 (20.76)	0.8035 (20.41)
15	1.0605 (26.94)	1.0485 (26.63)	1.0265 (26.07)	1.0165 (25.82)	0.9885 (25.11)	0.9705 (24.65)
23	1.4980 (38.05)	1.4860 (37.74)	1.4640 (37.18)	1.4540 (36.93)	1.4260 (36.22)	1.4080 (35.76)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE A-5. Connector mating threads (external) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.



Inch	Millimeters
0.002	0.05
0.004	0.10
0.005	0.13
0.006	0.15
0.035	0.89
0.107	2.71
0.232	5.89
0.515	13.08

FIGURE A-6. Connector shell back-end configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.

Shell size	AA BSC	CC	V thread, class 2A	EE dia	FF dia	HH dia
11	20 degrees	0.0534 (1.356) 0.0484 (1.229)	0.750-20 UNEF 1/	0.547 (13.89) 0.541 (13.74)	0.538 (13.67) 0.532 (13.51)	0.571 (14.50) 0.565 (14.35)
13	20 degrees	0.0634 (1.610) 0.0584 (1.483)	0.875-20 UNEF	0.662 (16.81) 0.656 (16.66)	0.653 (16.59) 0.647 (16.43)	0.686 (17.42) 0.680 (17.27)
15	18 degrees	0.0679 (1.725) 0.0629 (1.598)	1.000-20 UNEF	0.797 (20.24) 0.791 (20.09)	0.788 (20.02) 0.782 (19.86)	0.821 (20.85) 0.815 (20.70)
23	12 degrees	0.0664 (1.687) 0.0649 (1.648)	1.4375-18 UNEF	1.224 (31.09) 1.222 (31.04)	1.217 (30.91) 1.211 (30.76)	1.250 (31.75) 1.244 (31.60)

FOOTNOTE:

1/ Unified Extra Fine Thread (UNEF).

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Dimensions apply to plated/finished part.
5. This design information establishes interface criteria for backshell attachment in accordance with figure A-7.
6. When properly seated, the rear of the connector insert will be 0.272±0.020 inch from rear of connector shell.
7. Minimum inside diameter of connector shell back end shall be 0.412 inch (10.46 millimeters) for shell size 11, 0.534 inch (13.56 millimeters) for shell size 13, 0.712 inch (18.08 millimeters) for shell size 15, and 1.118 inch (28.40 millimeters) for shell size 23. Minimum diameters apply to a depth of 0.300 inch (7.62 millimeters).

FIGURE A-6. Connector shell back-end configuration – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.

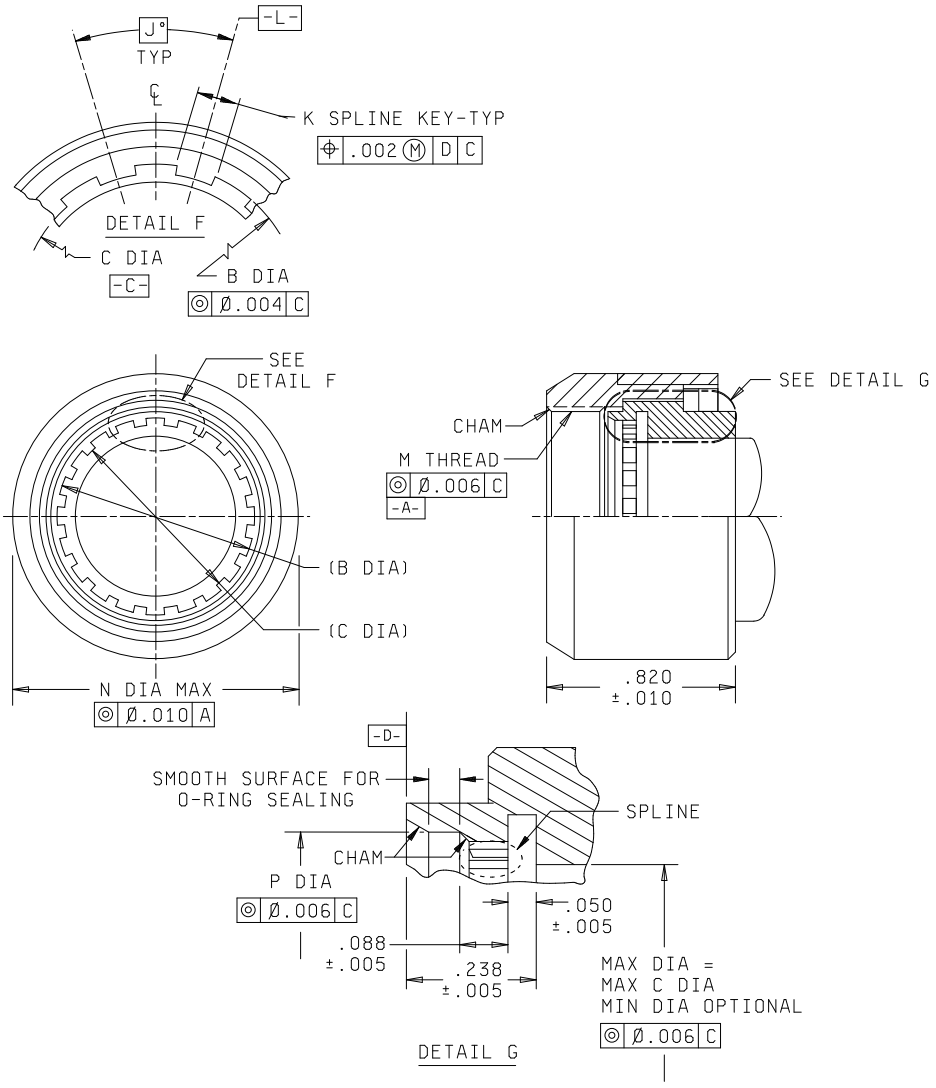


FIGURE A-7. Interface dimensions, connector, fiber optic backshell attachment nut.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.

Inch	Millimeters
0.002	0.05
0.004	0.10
0.005	0.13
0.006	0.15
0.010	0.25
0.050	1.27
0.088	2.24
0.238	6.05
0.820	20.83

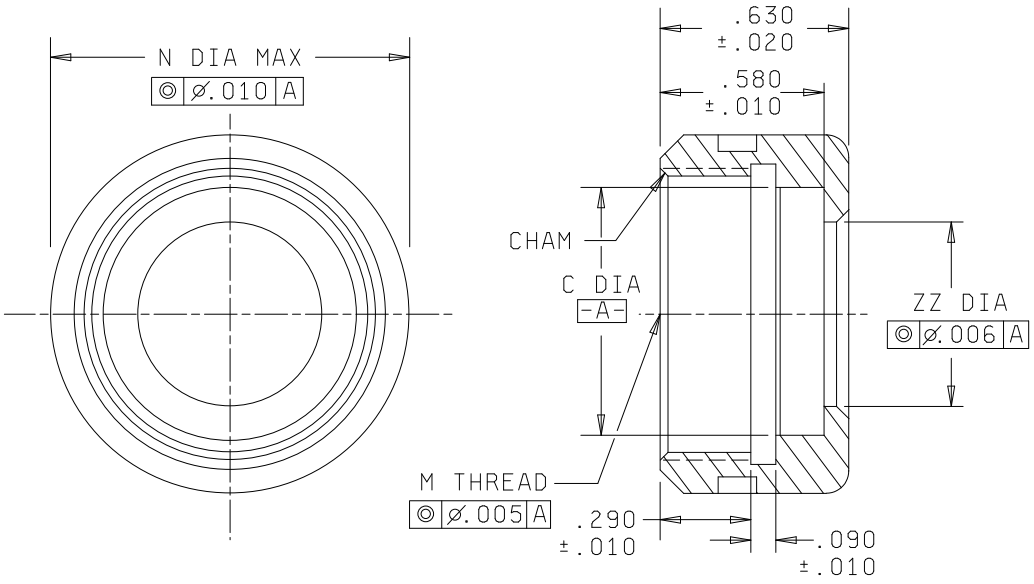
Shell size	B dia	C dia	J BSC	K	M thread, class 2B	N dia (max)	P dia
11	0.585 (14.86) 0.579 (14.71)	0.559 (14.20) 0.553 (14.05)	20 degrees	0.0445 (1.130) 0.0395 (1.003)	0.750-20 UNEF	0.960 (24.38)	0.617 (15.57) 0.605 (15.37)
13	0.700 (17.78) 0.694 (17.63)	0.674(17.12) 0.668 (16.97)	20 degrees	0.0545 (1.384) 0.0495 (1.257)	0.875-20 UNEF	1.085 (27.56)	0.732 (18.59) 0.720 (18.29)
15	0.835 (21.21) 0.829 (21.06)	0.809 (20.55) 0.803 (20.40)	18 degrees	0.0590 (1.499) 0.0540 (1.372)	1.000-20 UNEF	1.255 (31.88)	0.867 (22.02) 0.855 (21.72)
23	1.264 (32.11) 1.258 (31.95)	1.238 (31.45) 1.232 (31.29)	12 degrees	0.0575 (1.461) 0.0560 (1.422)	1.4375-18 UNEF	1.695 (43.05)	1.296 (32.92) 1.284 (32.61)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Dimensions apply to plated/finished part.
5. The coupling nut shall be captive and free to rotate on backshell.
6. Backshell insert retention hardware (e.g., insert retention sleeve) not shown. The backshell shall prevent the connector insert from moving rearward during connector mating. See figure A-6 for proper rear position of connector insert.
7. K dimension is the width of the spline key.

FIGURE A-7. Interface dimensions, connector, fiber optic backshell attachment – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.



Inch	Millimeters
0.005	0.13
0.006	0.15
0.010	0.25
0.020	0.51
0.090	2.29
0.290	7.37
0.580	14.73
0.630	16.00

FIGURE A-8. Interface dimensions, connector, fiber optic insert retention nut attachment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.

Shell size	M thread, class 2B	C dia	N dia (max)	ZZ dia
11	0.750-20 UNEF	0.616 (15.65) 0.594 (15.09)	0.960 (24.38)	0.340 (8.64)
13	875-20 UNEF	0.731 (18.57) 0.709 (18.01)	1.085 (27.56)	0.460 (11.68)
15	1.000-20 UNEF	0.866 (22.00) 0.844 (21.44)	1.257 (31.93)	0.645 (16.38)
23	1.4375-18 UNEF	1.306 (33.17) 1.284 (32.61)	1.763 (43.78)	1.065 (27.05)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Dimensions apply to plated/finished part.
5. Insert retention nut hardware (insert retention sleeve) not shown. The insert retention nut shall prevent the connector insert from moving rearward during connector mating. See figure A-1 and A-2 for proper rear position of connector insert.
6. Minimum inner diameter shall be 0.34 inch (8.636 millimeters) for shell size 11, 0.457 inch (11.608 millimeters) for shell size 13, 0.642 inch (16.307 millimeters) for shell size 15, and 1.042 inch (26.467 millimeters) for shell size 23.

FIGURE A-8. Interface dimensions, connector, fiber optic insert retention nut attachment – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.

CONNECTOR INSERT ARRANGEMENTS AND INTERFACE DIMENSIONS

B.1 SCOPE

B.1.1 Scope. This appendix lists the connector insert arrangements and interface dimensions. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

B.2 APPLICABLE DOCUMENTS

B.2.1 General. The document listed in this section are specified in Appendix B of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in Appendix B of this specification, whether or not they are listed.

B.2.2 Non-Government publications. The following document forms a part of this appendix to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Y14.5M - Dimensioning and Tolerancing.

(Copies of this document are available online at www.asme.org.)

B.3 REQUIREMENTS

B.3.1 Dimensions. Dimensions shall be in accordance with figures B-1 through B-6 herein, and the following dimensional data:

a. "▼" indicates the insert indexing feature position and vertical centerline of insert arrangement.

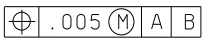
b. Dimensioning and tolerancing in accordance with ASME Y14.5M. (Dimensions are true position and are in inches.)

c. Metric equivalents are given in parentheses for general information only.

d. Dimensions, features, and markings shown are for engaging face of pin insert and the engaging face of the socket insert.

e. The following tolerance applies to an insert installed in a shell:

(1) The center of each hole in an insert for epoxy terminus connectors shall be located:



Where datum A is the connector shell master key/keyway

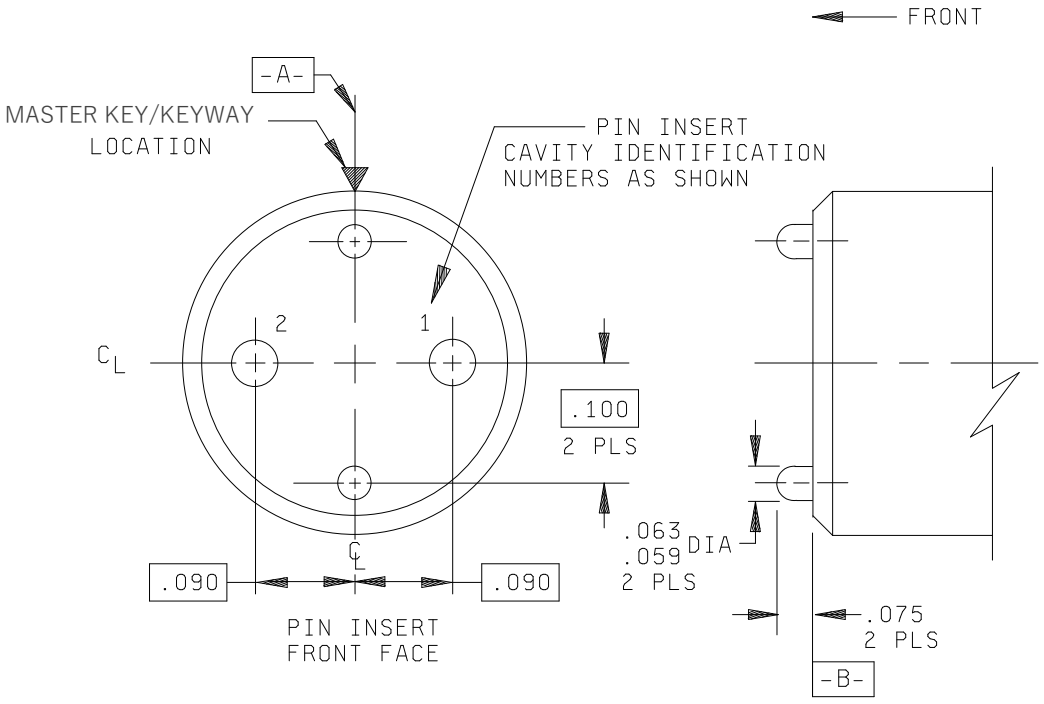
f. Unless otherwise indicated, dimensions are symmetrical about centerlines.

g. Each insert arrangement is shown in the "normal position" with indexing feature at top of vertical centerline of the engagement face.

h. Shell polarization shall be in accordance with figure A-3.

i. Tolerance is ±0.010 for three decimal places and ±0.030 for two decimal places unless otherwise stated.

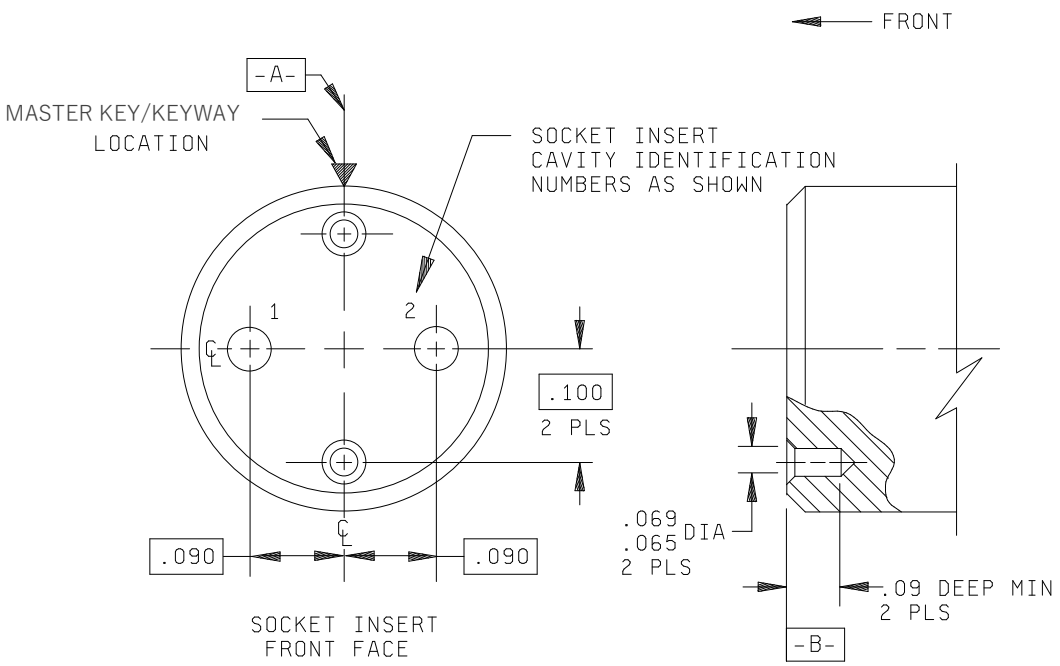
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.



Inch	Millimeters
0.059	1.50
0.063	1.60
0.065	1.65
0.069	1.75
0.075	1.91
0.090	2.29
0.100	2.54

FIGURE B-1. Two-position termini arrangement for shell size 11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.

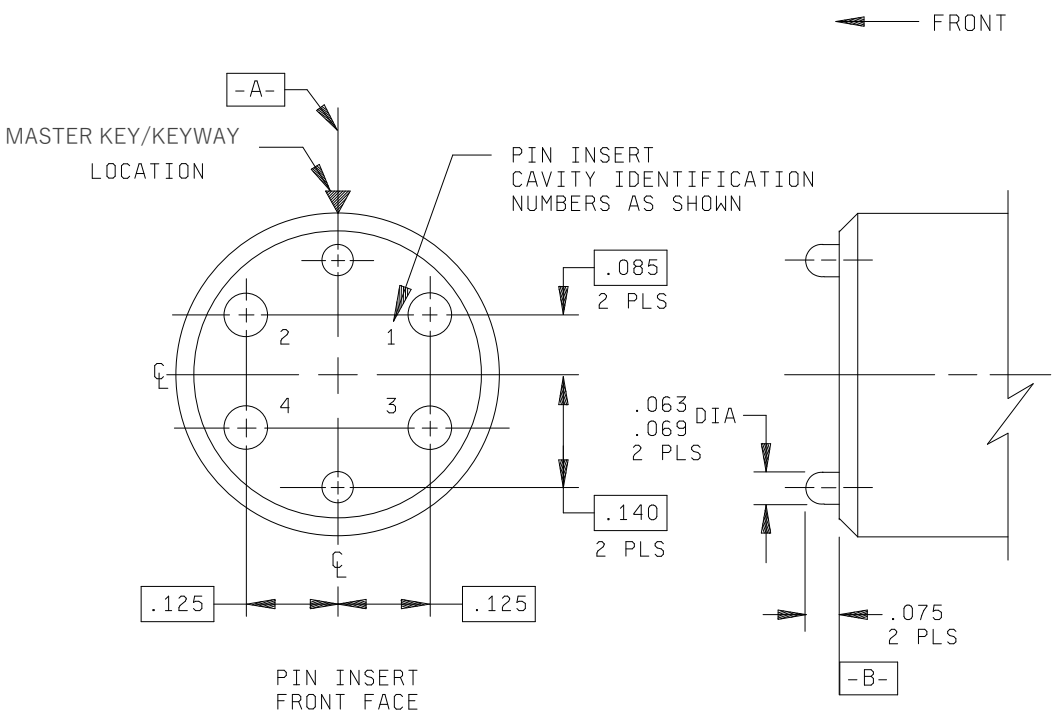


Shell size	Shell size designator	Arrangement number	Number of termini
11	A	1	2

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE B-1. Two-position termini arrangement for shell size 11 – Continued.

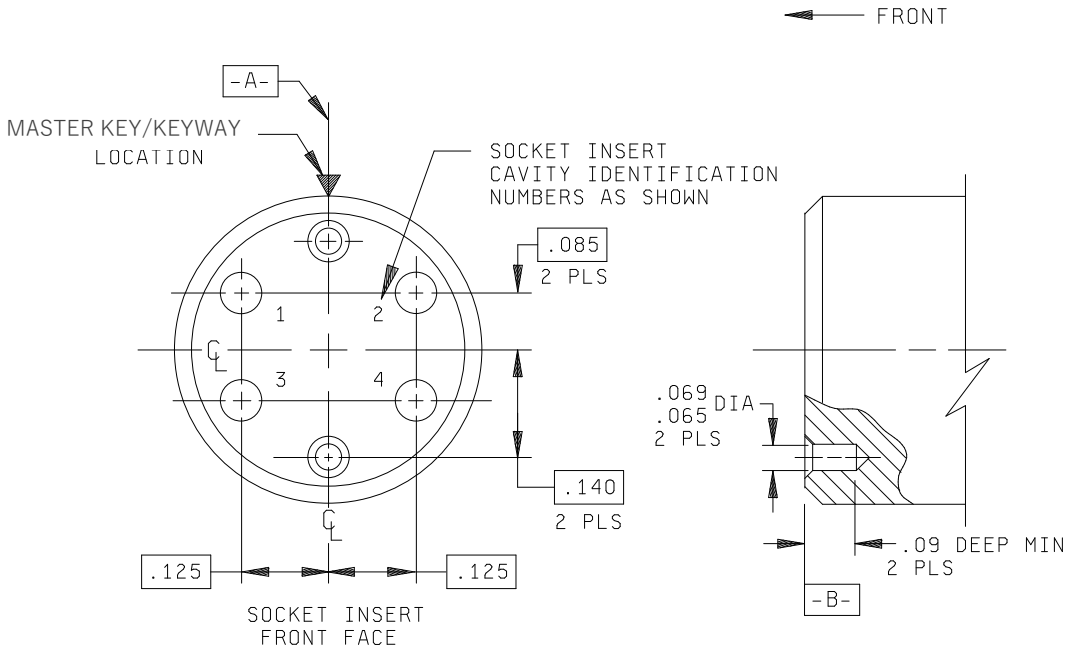
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.



Inch	Millimeters
0.063	1.60
0.065	1.65
0.069	1.75
0.075	1.90
0.085	2.16
0.090	2.29
0.125	3.18
0.140	3.56

FIGURE B-2. Four-position termini arrangement for shell size 13.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.

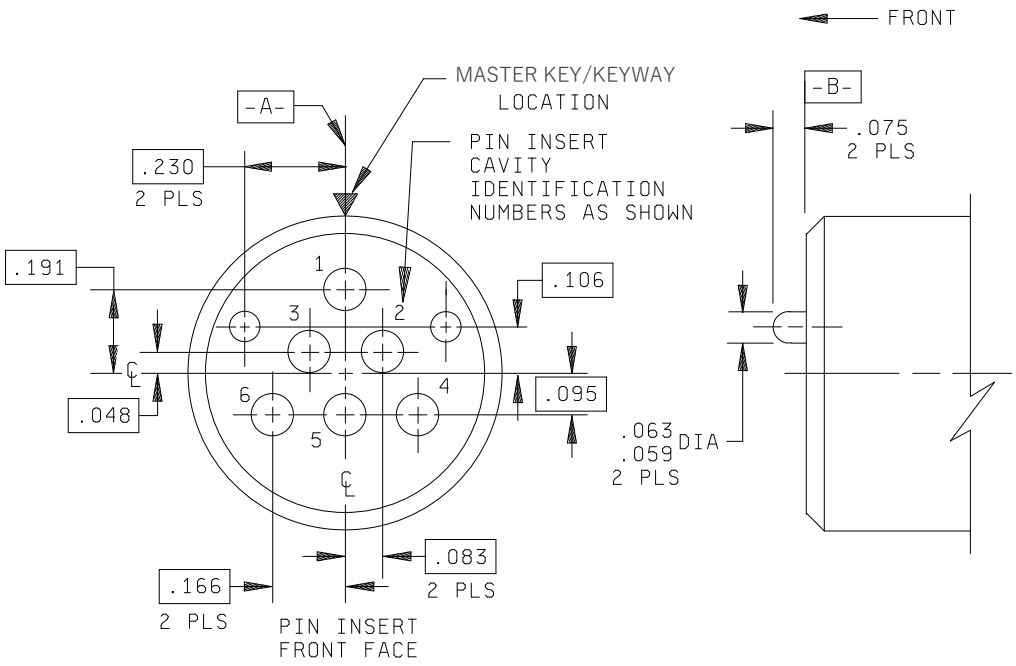


Shell size	Shell size designator	Arrangement number	Number of termini
13	B	1	4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE B-2. Four-position termini arrangement for shell size 13 – Continued.

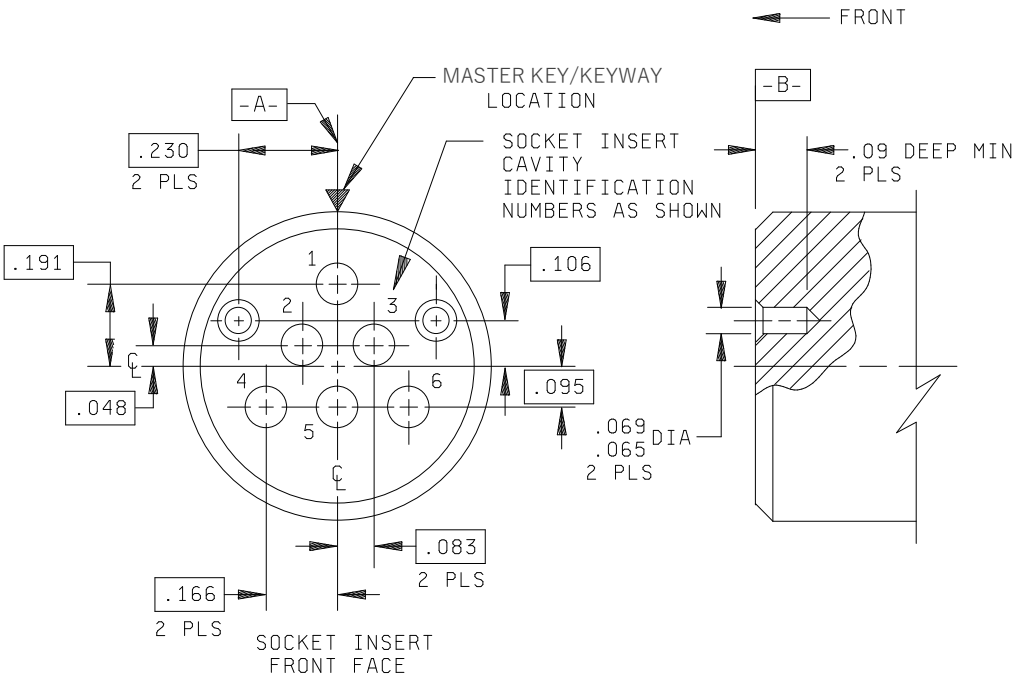
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
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Inch	Millimeters
0.048	1.22
0.059	1.50
0.063	1.60
0.065	1.65
0.069	1.75
0.075	1.90
0.083	2.11
0.090	2.29
0.095	2.41
0.106	2.69
0.166	4.22
0.191	4.85
0.230	5.84

FIGURE B-3. Six-position termini arrangement for shell size 15.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
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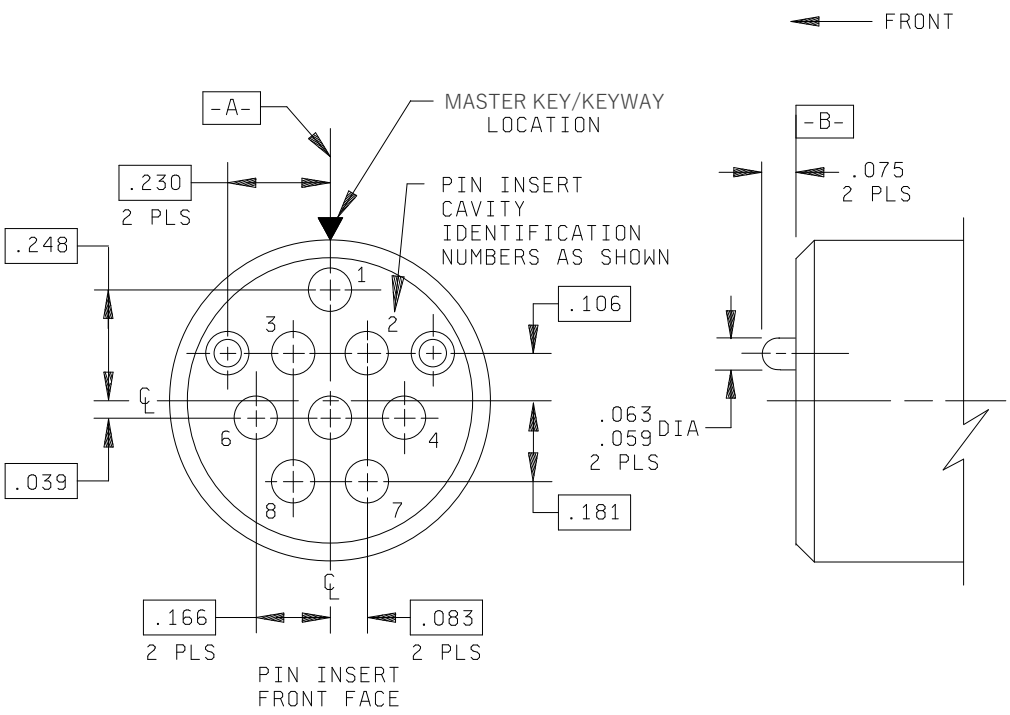


Shell size	Shell size designator	Arrangement number	Number of termini
15	C	2	6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE B-3. Six-position termini arrangement for shell size 15 – Continued.

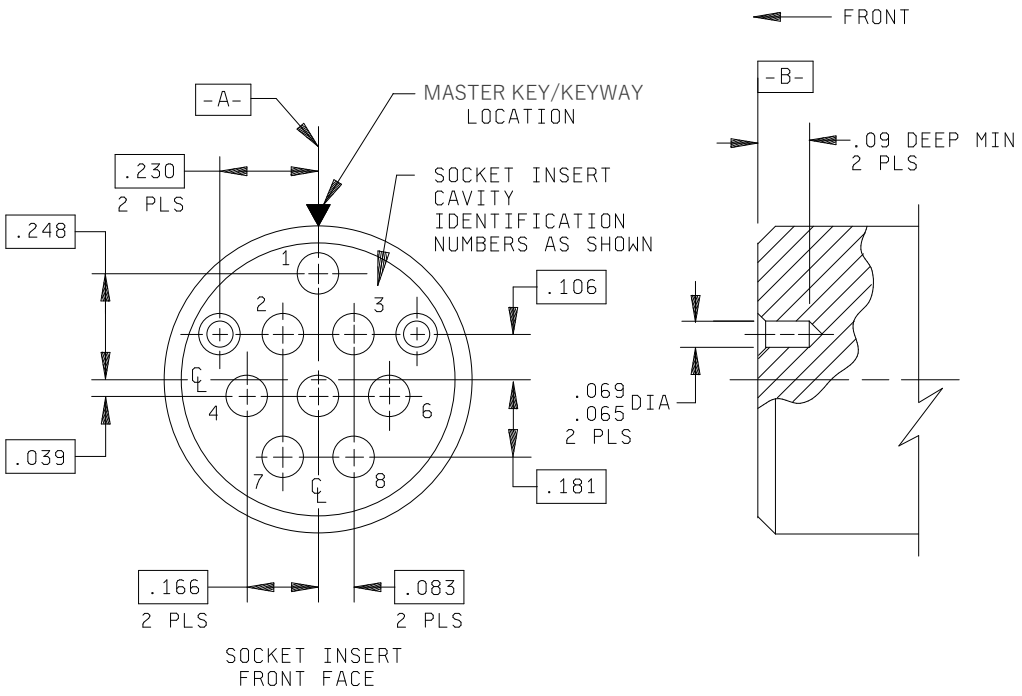
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.



Inch	Millimeters
0.039	0.99
0.059	1.50
0.063	1.60
0.065	1.65
0.069	1.75
0.075	1.90
0.083	2.11
0.090	2.29
0.106	2.69
0.166	4.22
0.181	4.60
0.230	5.84
0.248	6.30

FIGURE B-4. Eight-position termini arrangement for shell size 15.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.

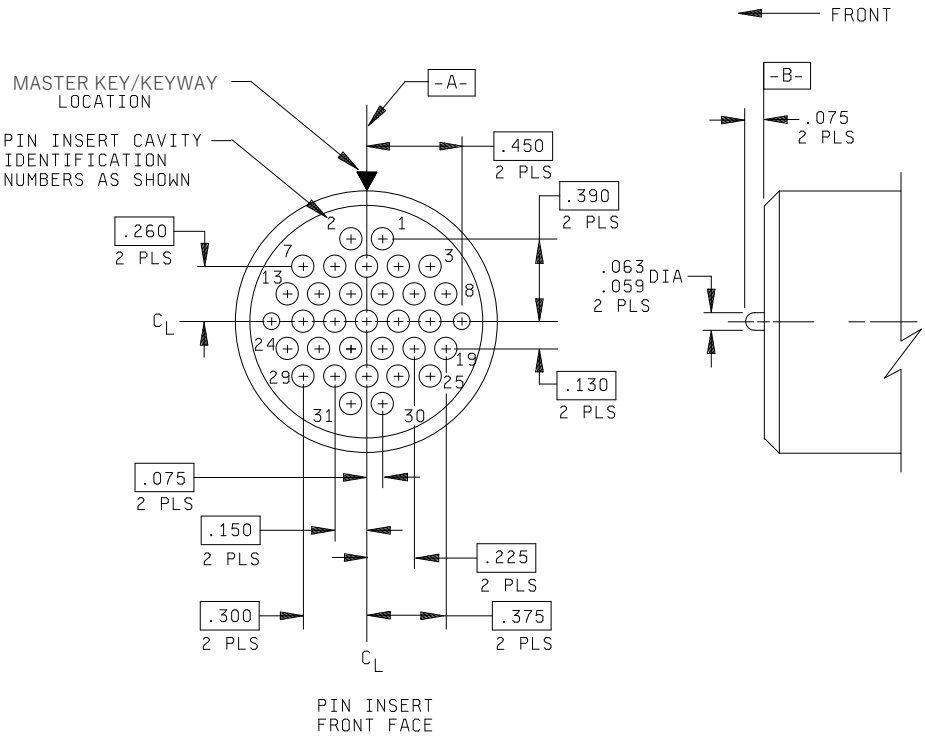


Shell size	Shell size designator	Arrangement number	Number of termini
15	C	1	8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE B-4. Eight-position termini arrangement for shell size 15 – Continued.

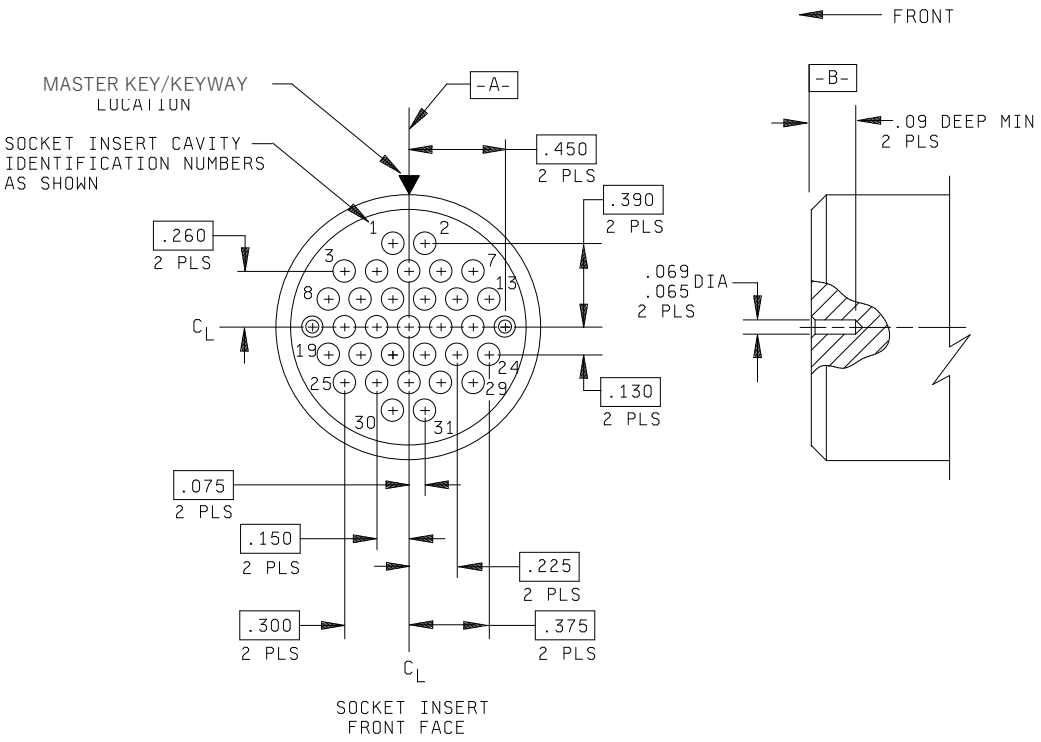
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.



Inch	Millimeters
0.059	1.50
0.063	1.60
0.065	1.65
0.069	1.75
0.075	1.90
0.090	2.29
0.130	3.30
0.150	3.81
0.225	5.72
0.260	6.60
0.300	7.62
0.375	9.53
0.390	9.91
0.450	11.43

FIGURE B-5. Thirty-one-position termini arrangement for shell size 23.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
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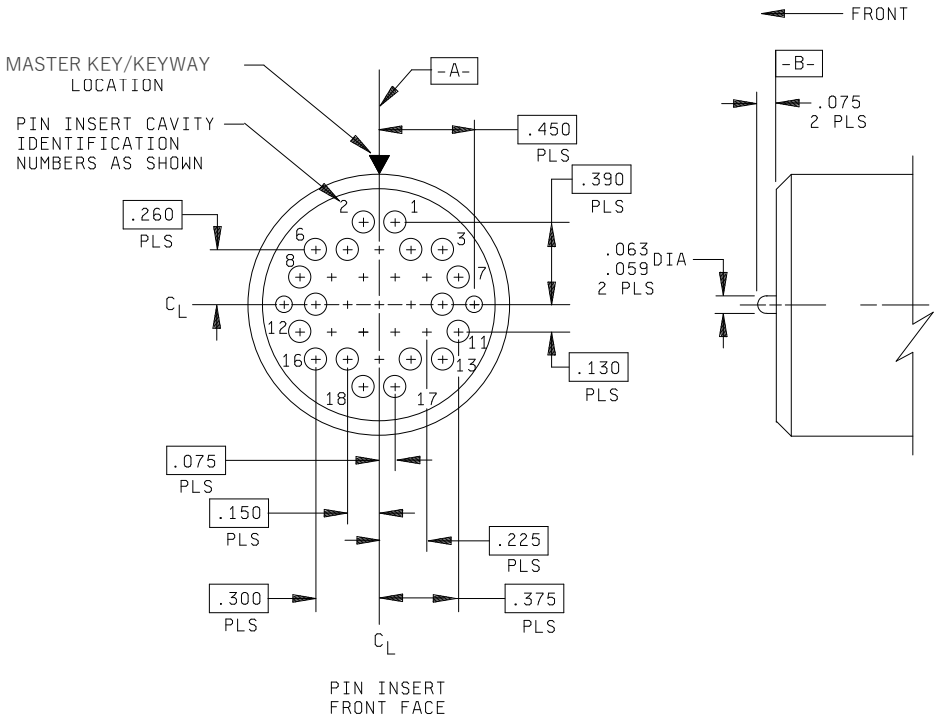


Shell size	Shell size designator	Arrangement number	Number of termini
23	F	1	31

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE B-5. Thirty-one-position termini arrangement for shell size 23 – Continued.

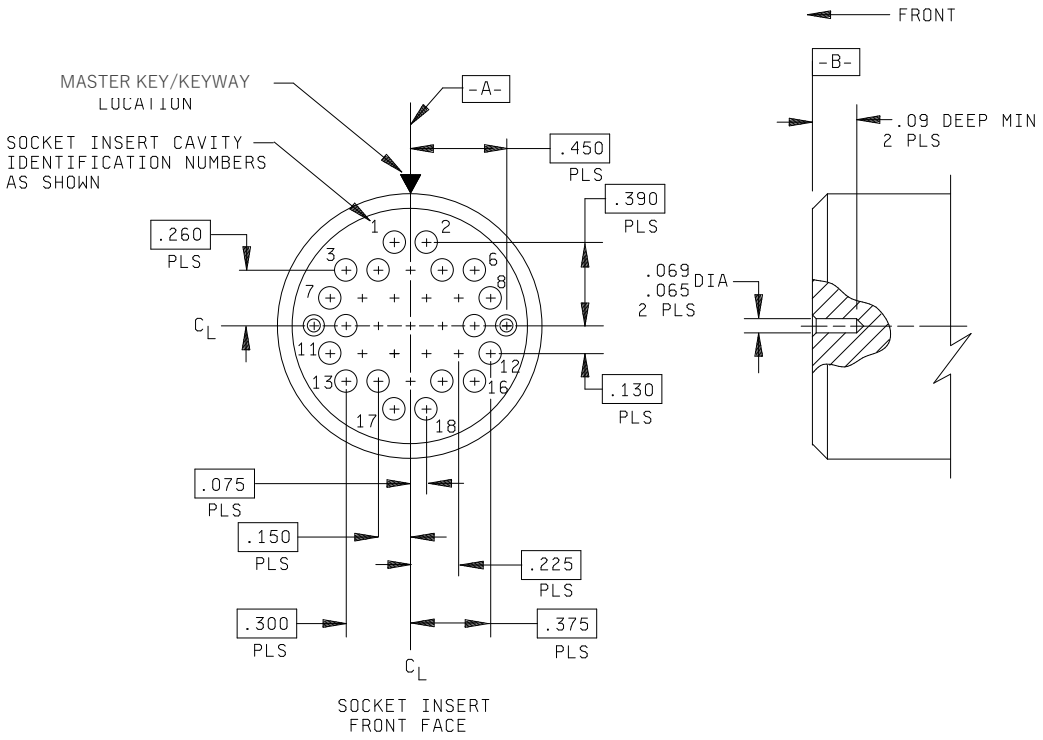
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:24Z
Check the source to verify that this is the current version before use.



Inch	Millimeters
0.059	1.50
0.063	1.60
0.065	1.65
0.069	1.75
0.075	1.90
0.090	2.29
0.130	3.30
0.150	3.81
0.225	5.72
0.260	6.60
0.300	7.62
0.375	9.53
0.390	9.91
0.450	11.43

FIGURE B-6. Eighteen-position termini arrangement for shell size 23.

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Shell size	Shell size designator	Arrangement number	Number of termini
23	F	2	18

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE B-6. Eighteen-position termini arrangement for shell size 23 – Continued.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA – CC

Preparing activity:
Navy – SH
(Project 6060-2014-023)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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